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Enhancing Korean Automated Essay Scoring via Linguistically Informed Augmentation and Topic-Aware Preprocessing

Ganghee Go¹ , Doguk Kim² 

¹Department of Electrical and Computer Engineering, Inha University, Korea

²Department of Artificial Intelligence, Inha University, Korea

Corresponding author: Doguk Kim; Email: dgkim@inha.ac.kr

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Abstract

Automated Essay Scoring (AES) for Korean texts faces significant challenges due to limited annotated data and the morphological complexity of the Korean language. In this study, we apply and evaluate two linguistically informed augmentation methods—BART-based CHEF and ELECTRA-based Random Masking Insertion (RMI)—tailored to the characteristics of Korean. Additionally, we adopt a topic-aware preprocessing strategy that embeds essay topics into model inputs to enhance topic awareness during training. Experiments on 44,658 Korean essays with 10-fold prompt-stratified cross-validation show that both augmentation methods and topic-aware preprocessing yield statistically significant improvements over a KoBERT-based baseline (paired *t*-tests, all surviving a Holm correction), and that combining them raises the quadratic weighted kappa (QWK) for the final composite score from 0.4243 to 0.4360, with no significant degradation on any rubric item. To contextualize these scores, we recover the three raters’ original scores and measure human agreement under the identical protocol. Model performance closely tracks the human agreement profile across rubric items (Spearman $\rho = 0.82$), and the headline QWK reaches 83% of the human ceiling of 0.523, measured as the agreement of one rater with the average of the other two—the same target the model predicts. These findings highlight the potential of linguistically informed augmentation and topic-aware preprocessing in building robust Korean AES systems.

Keywords: automated essay scoring; text data augmentation; topic-aware preprocessing; korean morphemes; inter-annotator agreement

1. Introduction

Recent educational reforms in South Korea have increasingly emphasized the cultivation of creativity and independent thinking among students. In alignment with these initiatives, national assessment formats are undergoing a transition from conventional multiple-choice tests to more expressive, open-ended formats such as narrative and essay-based evaluations. These formats are designed to better assess students’ higher-order cognitive and interpretive abilities. However, the inherently subjective nature of open-ended responses introduces challenges in achieving consistency and reliability in scoring. In response, there has been a growing interest in developing AES systems tailored to the Korean language.

A significant milestone in this domain was the release of a large collection of annotated essays—referred to as the *Essay Writing Evaluation Data*^a—by *AI-HUB*^b, under the supervision of the *National Information Society Agency (NIA)*. Although the official catalogue advertises roughly 50,400 essays, the actually distributed raw data contains 45,497 labeled essays; after removing duplicates, our study uses 44,658 essays (see Section 3 for the exact preprocessing pipeline). This corpus has become a foundational resource for Korean AES research (Lee and Kim 2023; Cho et al. 2021; Park and Lee 2022). Despite its contributions, the dataset remains limited in scale for training robust neural models. Although it comprises 44,658 essays across 50 distinct prompts, each prompt includes only approximately 900 samples, resulting in sparse topic-level representation. Furthermore, no comparable large-scale Korean essay dataset is currently available, rendering data scarcity a significant bottleneck for AES development.

Data augmentation presents a viable strategy to mitigate these limitations. However, its application in the context of Korean AES remains relatively underexplored. Easy Data Augmentation (EDA) (Wei and Zou 2019), one of the most widely adopted techniques in Korean NLP, consists of four operations: synonym replacement (SR), random insertion (RI), random swap (RS), and random deletion (RD). While EDA is computationally efficient and easy to implement, it often fails to preserve semantic and grammatical integrity, particularly in morphologically rich languages such as Korean (Seo and Oh 2023). In particular, synonym replacement in EDA lacks contextual sensitivity, frequently producing awkward or ungrammatical constructions due to disregard for syntactic dependencies.

To address these shortcomings, several alternative approaches have been proposed. These include embedding-based synonym replacement and Random Masking Replacement (RMR), in which selected tokens are replaced with [MASK] tokens and subsequently predicted using a pre-trained language model (Wu et al. 2019; Kobayashi 2018; Kim et al. 2020). Although these methods exhibit improved contextual coherence compared to rule-based techniques, they still encounter difficulties in handling Korean's agglutinative morphology. Mismatches at the morphological or syntactic level often lead to semantically or grammatically inappropriate outputs. Moreover, the effectiveness of these methods in AES-specific contexts remains largely unverified, as they have not been systematically evaluated within this domain.

More recently, large-scale language models (LLMs) have demonstrated considerable potential for data augmentation in low-resource settings, including Korean AES. Among these, ChatGPT has been the most studied; previous research (Kim and Park 2023) reported that ChatGPT-generated text exhibits high semantic fidelity and naturalness, as evidenced by cosine similarity metrics and improvements in the performance of the AES model. Nevertheless, the high computational and financial costs associated with API-based LLMs pose barriers to widespread adoption in academic research. In contrast, open-source LLMs such as LLaMA^c support local inference and fine-tuning, allowing for the development of augmentation strategies that are more linguistically and appropriately tailored for the task. However, these models demand substantial computational infrastructure, which may be inaccessible to smaller institutions. Consequently, while LLM-based approaches are promising, their practical applicability in Korean AES is limited by both resource and cost constraints. This underscores the need for augmentation methods that are computationally efficient, linguistically robust, and suitable for morphologically complex languages such as Korean.

To address these challenges, the present study adapts and empirically evaluates two linguistically informed data augmentation methods—CHEF and Random Masking Insertion (RMI)—which, although previously proposed in Korean NLP literature, have not yet been applied to AES tasks. Unlike LLM-based approaches, these methods leverage pretrained language models

^a<https://url.kr/qilx39>

^b<https://www.aihub.or.kr/>

^c<https://huggingface.co/meta-llama>

(PLMs), offering a more computationally efficient alternative that is better suited for resource-constrained environments. Moreover, both CHEF and RMI are designed to preserve semantic integrity while introducing meaningful linguistic variation. CHEF generates grammatically valid and semantically coherent variants by decomposing and recombining lexical morphemes, thereby leveraging the agglutinative nature of Korean (Seo et al. 2023). RMI extends the RMR approach by inserting, rather than replacing, masked tokens, thereby preserving the original lexical context while enabling controlled variation through masked language modeling (Yang 2022).

In addition, a topic-aware preprocessing strategy, inspired by prior work on prompt- and topic-aware scoring (Gupta 2023; Cho et al. 2021; Jiang et al. 2023), is adopted. This approach involves embedding prompt information into each sentence of an essay to improve topic sensitivity. Given that the *Essay Writing Evaluation Data* comprises 50 distinct prompts—comparable in structure to the ASAP^d dataset used in English AES—the original prompt-insertion method is adapted to account for the relatively short sentence lengths in Korean essays. In this modified version, each sentence is explicitly labeled with the corresponding topic. We emphasize that topic-aware scoring itself is an established technique; our contribution lies in its adaptation to a Korean AES model and in a rigorous, statistically validated measurement of its effect.

Ultimately, the contributions of this study are summarized as follows:

1. We present the first systematic application of two linguistically informed augmentation techniques—CHEF and RMI—to Korean AES, together with a topic-aware preprocessing strategy adapted to Korean (word-boundary insertion for RMI, sentence-level topic labeling).
2. We show that every proposed component yields a statistically significant improvement over the baseline: under 10-fold prompt-stratified cross-validation with paired *t*-tests, CHEF, RMI, and topic-aware preprocessing each improve performance ($p = 0.0044$, 0.0106 and 0.0008 ; all three survive a Holm correction over the five confirmatory comparisons), with no significant degradation on any rubric item. Augmentation on top of topic-aware preprocessing is positive in both cases and yields our best configuration, but is nominally rather than adjusted-significant, and we report it as suggestive. As a supplementary comparison, we also situate the baseline against the official word2vec-based reference models distributed with the corpus and five traditional machine learning methods under the identical protocol.
3. We recover the three raters’ original scores and measure human agreement for all 11 rubric items, under the same protocol used to score the model. The per-item performance profile is tightly aligned with human agreement (Spearman $\rho = 0.82$), and the composite-score QWK of our best configuration (0.4360) reaches 83% of the human ceiling of 0.523—the agreement a single rater achieves with the average of the other two, which is precisely what the model is asked to predict.

2. Related Work

This section reviews previous work across four areas: (1) Automated Essay Scoring, (2) Korean Automated Essay Scoring, (3) general text data augmentation, and (4) Korean-specific text data augmentation. Through this review, we situate the present study within the broader literature and identify the relevance of adapting linguistically informed augmentation methods to Korean AES.

2.1 Automated Essay Scoring

AES has undergone significant evolution over the past two decades. Early systems relied on statistical models with manually engineered features (Attali and Burstein 2006; Phandi et al. 2015). Later, neural approaches using LSTM, CNN, and BiLSTM architectures enabled automatic feature

^d<https://paperswithcode.com/dataset/asap>

learning from raw text (Alikaniotis et al. 2016; Taghipour and Ng 2016; Dong and Zhang 2016). The introduction of attention mechanisms and sentence-level representations further improved models' ability to capture discourse-level coherence (Dong et al. 2017; Dasgupta et al. 2018), and several studies have modeled local coherence and discourse structure to enhance scoring fidelity (Farag et al. 2018; Tay et al. 2018; Li et al. 2018; Mesgar and Strube 2018). A recurring concern throughout this progression is that neural scorers latch onto superficial correlates of quality rather than quality itself. Essay length is the standard example: it correlates strongly with human scores, is trivial to compute, and a model that leans on it can post competitive agreement while being blind to content. Jeon and Strube (2021) therefore proposed scoring from content representations that are explicitly decorrelated from such surface cues. This concern is directly relevant to the present work, because the corpus we use exhibits precisely this correlation between length and school level (Section 3.1), and because two of its rubric items reward length and sentence form explicitly.

Trait-based scoring—predicting the individual rubric dimensions rather than a single holistic score, as we do here—forms a second relevant strand. Ridley et al. (2021) introduced a multi-task model in which a trait-attention mechanism lets each trait attend to different parts of the essay while sharing a common representation, reporting strong holistic and trait-level results on ASAP and ASAP++^e; Kumar et al. (2022) showed that a simpler multi-task BiLSTM can learn global quality and fine-grained traits jointly, at the cost of trait-specific modelling. Both findings motivate our choice to predict all 11 rubric items from a single shared essay representation with one output unit per item.

More recently, pre-trained language models (PLMs) such as BERT, XLNet, and Sentence-BERT have been integrated into AES pipelines (Yang et al. 2020; Wang et al. 2022; Li et al. 2023), alongside hybrid models that combine neural representations with hand-crafted features (Uto et al. 2020) and contrastive learning approaches (Xie et al. 2022). With the emergence of large language models (LLMs) like GPT-4 and ChatGPT, the AES landscape has broadened to include feedback generation (Stahl et al. 2024), second-language writing assessment (Yancey et al. 2023), human-AI collaborative scoring (Xiao et al. 2025), and interpretable zero-shot trait-level scoring (Lee et al. 2024).

Of particular relevance to this study is the line of work on prompt- and topic-aware scoring. Jiang et al. (2023) proposed disentangled representation learning to separate prompt-specific and prompt-invariant components of essay representations, improving domain generalization for prompt-aware scoring; Gupta (2023) showed that directly injecting prompt information into model inputs improves AES performance. These studies establish that making the scoring model aware of the prompt is beneficial; our work adapts this idea to a Korean AES model whose encoder is frozen—so that the prompt has to be supplied to the scoring module rather than to the encoder—and quantifies its contribution with formal significance testing.

2.2 Korean Automated Essay Scoring

Early work in Korean AES emphasized rule-based NLP approaches tailored to Korean linguistic characteristics such as free word order and frequent ellipsis. Noh et al. (2014) proposed a sentence-level scoring system based on morphological, syntactic, and semantic role analyses. The release of the *Essay Writing Evaluation Dataset* by AI-HUB in 2021 was a major milestone, enabling the development of baseline models and spurring subsequent research. Recent studies have explored a range of architectures and evaluation targets. For instance, Cho et al. (2021) employed KoBERT and KoGPT2^f to build topic-aware scoring models on this corpus, demonstrating that jointly encoding the essay topic and text benefits Korean AES; our topic-aware preprocessing builds on

^e<https://lwsam.github.io/ASAP++/lrec2018.html>

^f<https://github.com/SKT-AI/KoGPT2>

this insight. GRU-based models have shown advantages over RNN and LSTM in both efficiency and accuracy (Park and Lee 2022), while tree-based methods like Random Forests have been valued for interpretability (im Park and ki Choi 2023). Advanced strategies, including structured representation learning and explainable AI, have also gained traction. Lee and Kim (2023) incorporated argument structure information to enhance scoring precision, and Lee (2025) proposed a hybrid model combining shallow linguistic features and semantic embeddings. The PASTA-I model, based on ELECTRA, was designed for rubric-aligned scoring (Lee et al. 2023), and Im and Lim (2024) evaluated ChatGPT-based AES in classroom settings.

A practical difficulty in this literature is that reported results are rarely directly comparable. Studies based on the *Essay Writing Evaluation Data* differ in prediction targets (holistic total score vs. individual rubric items), score scales and binning schemes, data subsets, and validation protocols, and several report accuracy- or correlation-based metrics rather than QWK. For instance, Lee and Kim (2023) fine-tune KLUE-RoBERTa on the argumentative subset of the corpus (20,442 training essays) and report a total-score QWK of 0.543 for their baseline and 0.627 for their argument-structure-aware model, where the total score is the sum of the eleven 0–3 item scores rounded to integers; Cho et al. (2021) instead frame the task as score-range classification and report accuracy. Neither setting is commensurable with our protocol, which uses the full corpus, frozen-encoder features, and a quartile-binned weighted composite score. As a consequence, no established QWK reference exists for the full corpus under a rubric-level protocol such as ours. We therefore contextualize our results in two ways: (i) as a supplementary experiment, we re-implement and evaluate, under an identical cross-validation protocol, the official word2vec-based LSTM/GRU reference models distributed with the corpus, together with five traditional machine learning methods (Section 5.3); and (ii) we measure the inter-annotator agreement of the corpus itself to establish a human agreement reference that contextualizes attainable QWK (Section 3.3). Despite these efforts, Korean AES still suffers from limited data resources. Most datasets are narrow in scope or small in scale, constraining model robustness and generalizability. This highlights the importance of exploring augmentation strategies suited to the morphological and stylistic richness of Korean.

2.3 Text Data Augmentation

Text data augmentation (DA) has become increasingly important for NLP, particularly in low-resource settings. Unlike continuous domains like computer vision, language's discrete nature makes it difficult to modify text without distorting meaning (Li et al. 2022). The methods proposed in response can be grouped by the mechanism through which they introduce variation, and the trade-off each mechanism makes between diversity and fidelity is what distinguishes them. Early DA methods include synonym replacement using WordNet (Zhang et al. 2015; Wei and Zou 2019), which swaps words for dictionary synonyms and therefore varies the surface form with no model of the surrounding context, and embedding-based substitutions (Wang and Yang 2015), which draw the replacement from the nearest neighbours of the original word in a distributional space and so respect topical fit but not syntactic agreement. More recent work has leveraged masked language models like BERT for context-aware perturbation (Wu et al. 2019; Kobayashi 2018; Kumar et al. 2020), in which a token is replaced by [MASK] and the model predicts a substitute conditioned on the entire remaining sentence—removing the context-insensitivity of the two earlier families at the cost of requiring a pretrained model—and back-translation for meaning preservation (Sennrich et al. 2016), where a sentence is translated into a pivot language and back, producing a fluent paraphrase over whose content the practitioner has little control. Noising techniques such as insertion, deletion, and swap (Wei and Zou 2019) provide lightweight alternatives that require no model at all, but degrade grammaticality roughly in proportion to how much they perturb. Beyond these, generative and self-training approaches synthesize new examples outright rather than editing existing ones: template- and rule-based sampling (Kang et al. 2018), conditional generation

from a fine-tuned language model (Anaby-Tavor et al. 2020), interpolation of hidden representations across examples (Chen et al. 2020), and model-in-the-loop pair construction for sentence encoders (Thakur et al. 2021). Most recently, LLMs enable instruction-driven generation (Li et al. 2024; Whitehouse et al. 2023), denoising (Ghosh et al. 2023), synthetic labeling (Xiao et al. 2023), and teacher-student distillation pipelines (Khattab et al. 2024; Viswanathan et al. 2023), at correspondingly higher computational or API cost. In the context of AES, DA has been used to mitigate topic imbalance and data scarcity. For example, Jong et al. (2022) applied back-translation with score adjustments, Park et al. (2022) proposed EssayGAN for score-controlled essay generation, topic-aware augmentation using BART summarization was introduced by Gupta (2023), and rubric-specific augmentation has been studied by Yoo et al. (2025).

2.4 Korean Text Data Augmentation

In Korean NLP, DA has been explored to alleviate resource constraints. Early efforts adapted English-based methods—EDA, Masked Language Modeling (MLM), and back-translation—to Korean tasks (Vu et al. 2022; Kim et al. 2020; Seo and Oh 2023). Vu et al. (2022) showed performance gains on STS and NLI tasks. Kim et al. (2020) used a BERT-based MLM to overcome rule-based limitations, while Seo and Oh (2023) introduced label-preserving strategies such as DA-SC (spelling correction), EDA-POS (POS-aware augmentation), and DA-cMLM (conditional masking), which improved results on small datasets. Subsequent work tailored augmentation more closely to Korean linguistic properties. Yang (2022) proposed Random Masking Replacement (RMR) and Random Masking Insertion (RMI): rather than replacing existing tokens—which in Korean frequently breaks morphological agreement between stems and particles—RMI inserts [MASK] tokens at sampled positions and lets a masked language model generate contextually appropriate insertions, preserving all original tokens. Their experiments showed that RMI better preserves meaning while improving downstream performance. Seo et al. (2023) proposed CHEF, which exploits Korean’s agglutinative morphology: a sentence is decomposed into lexical morphemes (content words) and grammatical morphemes (particles and endings), and a sequence-to-sequence blender is trained to reconstruct full sentences from the lexical morphemes only, yielding stylistically diverse but semantically consistent variants. Kim and Park (2023) explored using ChatGPT for Korean essay augmentation. The study found that generated texts retained semantic fidelity and reduced manual workload, offering a relatively efficient alternative to human-authored data. However, reliance on API-based access limits its scalability and cost-effectiveness in long-term academic settings.

While these studies highlight the potential of data augmentation tailored to Korean, their application to complex tasks like AES remains limited. In particular, AES presents unique challenges due to its linguistic complexity and topic sparsity, underscoring the need for empirical investigations into how existing augmentation techniques can be effectively adapted to Korean-specific contexts. The present study fills this gap by evaluating CHEF and RMI—the two methods above that are explicitly designed around Korean morphology—on the largest public Korean essay corpus with a statistically rigorous protocol.

3. Dataset

This section describes the corpus, its rubric structure, its descriptive statistics, and its annotation quality. We present the dataset before the methodology so that the rubric structure referenced by the model architecture and the evaluation protocol is established first.

Table 1. Composition of the corpus used in this study (after deduplication).

Item	Value
Essays	44,658 (45,497 distributed raw essays – duplicates)
Prompts (topics)	50
School level	Elementary 17,319 (G4 5,246 / G5 6,085 / G6 5,988); Middle 17,167 (G7 6,025 / G8 5,832 / G9 5,310); High 10,172 (G10 4,273 / G11 4,335 / G12 1,564)
Writing mode	Analytical (<i>nonsul</i>) 23,350; Descriptive (<i>seosul</i>) 21,308
Genre	Expository 16,958; Argumentative 9,876; Opinion (pro/con) 7,535; Proposal 5,939; Creative 4,350
Scoring	11 rubric items (0–3 each), average of three human raters
Total score	25.08 ± 3.83 (range 0–30)

Table 2. Essay length statistics (mean ± std, median, min–max), measured on the full corpus. Minima and maxima are taken per column and therefore need not refer to the same essay.

Group	n	Sentences	Words (eojeol)	Characters (no spaces)
All	44,658	11.5 ± 5.1 (med 11, 1–97)	140.5 ± 62.2 (med 135, 6–1,207)	449.8 ± 195.0 (med 418, 138–3,816)
Elementary	17,319	10.0 ± 4.9 (med 9, 1–62)	96.4 ± 39.8 (med 88, 10–364)	316.5 ± 126.4 (med 288, 138–1,161)
Middle	17,167	11.4 ± 3.9 (med 11, 1–54)	149.1 ± 38.9 (med 145, 6–662)	467.7 ± 122.6 (med 452, 155–1,577)
High	10,172	14.3 ± 6.0 (med 14, 1–97)	201.0 ± 68.2 (med 202, 11–1,207)	646.5 ± 211.9 (med 651, 143–3,816)

3.1 Corpus Description

We use the *Essay Writing Evaluation Data*, the largest publicly available Korean essay corpus developed under the supervision of the *National Information Society Agency (NIA)*. The dataset contains essays written by students from grade 4 of elementary school to grade 12 of high school. The exact size of this corpus has been reported inconsistently in the literature; for example, [Misgna et al. \(2024\)](#) cite 50,413 essays, matching the official catalogue description of roughly 50,400. However, the actually distributed raw data contains 45,497 essays with complete rubric annotations. Starting from these 45,497 essays, we removed duplicate essays—pairs with identical essay content, identified both by exact string matching and by a sentence-embedding cosine similarity of 1.0 (i.e., textually indistinguishable duplicates)—yielding the final corpus of **44,658** essays used throughout this study. Table 1 summarizes its composition.

Table 2 reports essay length statistics measured over the entire corpus. Average essay length increases monotonically with school level (from 316.5 characters for elementary school to 646.5 for high school), indicating that length and grade level are strongly correlated. We return to the implications of this correlation for potential length-based shortcut learning in Section 6.

3.2 Rubric Structure and the Final Composite Score

Each essay is scored on a 0–3 scale for 11 rubric items, which are grouped into three broader categories: *Expression*, *Organization*, and *Content*. Table 3 describes the evaluation criteria for each rubric item. These scores refer exclusively to the *original* essays, and were produced entirely by the corpus distributors in 2021; we had no role in the annotation process. According to the corpus documentation, the 50 prompts were designed in consultation with domain experts, each essay was independently rated by three expert raters against the official rubric of eleven items, and records with missing or corrupted annotations were removed before release; the distributed label is the average of the three raters’ scores. Beyond this, the documentation discloses neither the raters’ professional qualifications and training, nor their compensation, nor any reliability statistics. Because the annotation procedure therefore cannot be characterized from the documentation alone, we measure its *outcome* directly: the raw distribution includes each of the three raters’ individual item scores, from which we compute inter-annotator agreement for every rubric

Table 3. Descriptions of the 11 rubric items used for essay evaluation.

Category	Rubric Item	Description
Expression	Grammar (exp1)	Accuracy of grammar and usage
	Vocabulary (exp2)	Lexical variety and appropriateness
	Sentence (exp3)	Syntactic complexity and fluency
Organization	Consistency (org1)	Logical flow and coherence
	Length (org2)	Adherence to expected length
	In-paragraph (org3)	Cohesion within paragraphs
	Structure (org4)	Presence of introduction, body, and conclusion
Content	Clarity (con1)	Clarity of thesis and ideas
	Novelty (con2)	Creativity and originality
	Prompt (con3)	Relevance to the given prompt
	Descriptions (con4)	Specificity and elaboration of content

item in Section 3.3. Augmented essays generated in this study (Section 4.2) inherit the rubric scores of their source essays and receive no additional human judgments; the validity of this label-preserving assumption is supported by the semantic preservation analyses in Section 5.2 and by the fact that augmented data are used only for training, never for evaluation (Section 5.1).

During preprocessing, raw rubric scores are linearly normalized to the range $[0, 1]$ for model training. In addition to trait-based evaluation (i.e., predicting scores for individual rubric items), we also compute a holistic score referred to as the *Final Composite Score (FCS)*. This metric aggregates rubric-level predictions using both item-specific and category-level weights.

The FCS is calculated as follows:

$$FCS = \sum_{c \in \{\text{exp, org, cont}\}} W_c \cdot \frac{\sum_{i \in \mathcal{I}_c} w_i \cdot (3 \cdot y_i)}{\sum_{i \in \mathcal{I}_c} w_i} \quad (1)$$

Here, $y_i \in [0, 1]$ denotes the normalized prediction for the i -th rubric item, w_i is its item-level weight, and $\mathcal{I}_c$ is the index set for category $c \in \{\text{Expression, Organization, Content}\}$. W_c is the category-level weight applied to each group. The item-level weights w_i and category-level weights W_c are not free parameters of our method: we use the official weighting scheme distributed with the corpus, which specifies rubric weights per essay type and is also used by the distributors to compute the weighted total score (0–30 scale) included in the raw data. The ground-truth FCS of an essay is obtained by applying Equation (1) to the three-rater average rubric scores with the same weights. Because the FCS is continuous, QWK on the FCS is computed after discretizing it into four ordinal classes. The binning is applied separately to each of the two score vectors being compared: within the set of essays under evaluation, each vector is cut at its own quartiles, so both sides become equal-frequency four-class labels before the quadratic weighted kappa is taken. In cross-validation this is done inside each validation fold, using only that fold's essays; no quantity derived from the training folds enters the binning, and no quantity derived from the ground truth is used to bin the predictions. The identical protocol is applied when measuring inter-annotator agreement on the FCS in Section 3.3, so that model and human scores are directly comparable. The complete weight table is given in Table 4.

Two properties of this scheme matter for the interpretation of our results. First, the weights are markedly non-uniform: *Consistency* (org1) and *Clarity* (con1) carry the largest item weight in most schemes, while *Sentence* (exp3), *Length* (org2) and *Prompt* (con3)—three of the four items on which both humans and models achieve near-ceiling agreement—receive a weight of 0 in four schemes each and therefore drop out of the composite score for the corresponding prompts. The FCS is consequently dominated by exactly the subjective items on which human agreement is weakest (Section 3.3), which is one reason why FCS QWK values on this corpus are much lower than the average rubric QWK. Second, because the weights differ across prompts, the composite score is not directly comparable across prompts; the quartile binning described above is applied to

Table 4. The eight rubric weighting schemes used by the corpus, covering all 50 prompts. Item weights w_i follow the order of Table 3; category weights W_c are given for Expression / Organization / Content. “Prompts” is the number of prompts assigned to each scheme.

Scheme	Prompts	Item weights w_i											Category weights W_c		
		exp1	exp2	exp3	org1	org2	org3	org4	con1	con2	con3	con4	Exp	Org	Con
S1	10	2	4	4	6	0	3	1	3	3	1	3	3	3	4
S2	4	2	4	4	3	3	3	1	3	3	1	3	3	3	4
S3	6	3	3	0	7	0	2	1	4	2	1	2	3	3	4
S4	8	2	3	1	4	2	2	1	4	3	0	1	4	2	4
S5	4	4	3	0	3	0	0	1	4	3	1	2	3	3	4
S6	8	4	3	0	5	0	0	1	4	4	0	2	4	2	4
S7	4	2	3	2	3	3	2	1	4	3	0	1	4	2	4
S8	6	3	3	0	5	1	1	1	4	4	0	2	4	2	4

the pooled composite scores, following the reference implementation for this corpus. The prompt-to-scheme assignment is summarized in the last column of Table 4.

3.3 Annotation Quality: Inter-Annotator Agreement

Essay grading is known to vary substantially between (and even within) human raters, and the reliability of the labels bounds what any model can achieve. To make this bound explicit, we obtained the raters’ original scores from the distributed raw data, which contain the individual rubric scores of the three raters (denoted a/b/c) for each essay. We verified the mapping against our working corpus by essay identifier: all 44,658 essays (100%) were matched, and the three-rater average reproduces the distributed label exactly for all 11 items. We then computed the inter-annotator agreement (IAA) for each rubric item as the pairwise quadratic weighted kappa (QWK)—the same metric used for model evaluation—over the full corpus.

Pairwise agreement alone, however, is not the quantity a model should be compared against. The model predicts the *average* of the three raters, a target whose noise is reduced by averaging, whereas a pairwise kappa compares one individual rater with another. To place a human rater on exactly the same footing as the model, we additionally report a **leave-one-out (LOO) agreement**: for each rater in turn, we score that rater’s judgments against the average of the remaining two, under the identical protocol used for the model (per item, both sides rounded; at the composite level, quartile binning as in Section 3.2). The LOO value is the matched human reference; the pairwise value is reported alongside it because it is the quantity conventionally called IAA.

Table 5 reveals two findings that are central to interpreting all subsequent results. First, human agreement varies enormously across rubric items: objectively checkable items such as *Length* (0.948), *Sentence* (0.946), and *Prompt* (0.934) show near-ceiling agreement, whereas subjective items such as *Novelty* (0.284), *Clarity* (0.339), and *Descriptions* (0.364) show weak agreement. This directly confirms, on our corpus, the observation that essay grading varies significantly between raters. Second, the per-item agreement profile almost perfectly predicts the per-item performance profile of our baseline model: across the 11 items, human agreement and baseline QWK correlate at Pearson $r = 0.943$ ($p = 1.3 \times 10^{-5}$; Spearman $\rho = 0.791$, $p = 0.0037$) using pairwise agreement, and at $r = 0.951$ ($p = 6.7 \times 10^{-6}$; $\rho = 0.818$, $p = 0.0021$) using the matched LOO agreement. Because the 11 items fall into two well-separated groups—four near-ceiling items around 0.95 and seven subjective items around 0.3–0.5—the Pearson coefficient is inflated by this clustering, and we regard the rank correlation as the more conservative summary; both are reported. Items on which the model scores poorly are precisely those on which human raters disagree, and on six items (org2, org3, org4, con1, con3, con4) the model’s QWK meets or exceeds the matched LOO agreement of a human rater. At the item level the two human references are

Table 5. Human agreement on the corpus ($N=44,658$) and the corresponding baseline model performance. *Pairwise* compares two individual raters (the conventional IAA); *LOO* compares each rater against the average of the other two, which is the target the model predicts and therefore the matched human reference. The FCS row uses the identical quartile-binning protocol as model evaluation.

Rubric item	Pairwise QWK (a–b / a–c / b–c)	Mean pairwise	Mean LOO	3-rater unanimity	Model QWK (baseline)
Grammar (exp1)	0.469 / 0.367 / 0.414	0.417	0.387	48.5%	0.258
Vocabulary (exp2)	0.469 / 0.448 / 0.499	0.472	0.462	53.7%	0.320
Sentence (exp3)	0.945 / 0.943 / 0.951	0.946	0.948	75.3%	0.935
Consistency (org1)	0.396 / 0.420 / 0.469	0.428	0.427	43.5%	0.348
Length (org2)	0.948 / 0.949 / 0.947	0.948	0.947	76.3%	0.947
In-paragraph (org3)	0.912 / 0.904 / 0.906	0.907	0.902	61.1%	0.912
Structure (org4)	0.552 / 0.338 / 0.432	0.440	0.473	51.6%	0.620
Clarity (con1)	0.346 / 0.276 / 0.396	0.339	0.326	44.0%	0.378
Novelty (con2)	0.349 / 0.251 / 0.253	0.284	0.295	30.2%	0.143
Prompt (con3)	0.939 / 0.927 / 0.936	0.934	0.931	70.5%	0.949
Descriptions (con4)	0.389 / 0.342 / 0.360	0.364	0.353	40.5%	0.448
Average (11 items)	—	0.589	0.586	54.1%	0.569
FCS (quartile 4-class)	0.408 / 0.402 / 0.530	0.447	0.523	—	0.424

nearly identical (mean 0.589 pairwise vs. 0.586 LOO), so the conclusions above do not depend on which one is used.

At the composite level, however, the two references differ substantially, and the distinction matters for how the headline number should be read. Pairwise inter-rater FCS QWK averages 0.447, but the matched LOO reference—a single rater scored against the average of the other two, exactly the target the model is trained and evaluated on—is **0.523** (range 0.457–0.564). The baseline model’s 0.4243 therefore corresponds to 81% of the human ceiling on this metric, not to parity with it. We report both values, and use the LOO figure of 0.523 as the reference throughout the paper.

Three caveats apply. First, the rater identifiers a/b/c are anonymous slots, and rater identity is not guaranteed to be consistent across essays; the reported kappas are therefore slot-level agreement estimates. Second, the agreement figures are computed over the pooled corpus spanning all grade levels and prompts; pooling heterogeneous groups widens the marginal score distributions and can inflate QWK relative to within-group agreement, so the reference values above are, if anything, optimistic. Third, intra-rater consistency cannot be assessed from the distributed data, which we note as a limitation of the corpus. The asymmetry between the model’s target and a pairwise human comparison, which would otherwise be the most serious objection to this analysis, is removed by construction through the LOO reference.

4. Methodology

This section presents the model architecture, the two linguistically informed data augmentation methods (CHEF and RMI), and the topic-aware preprocessing strategy. The base architecture follows a conventional structure—sentence-level embeddings from a frozen KoBERT[§] processed by a GRU (Gated Recurrent Unit)-based scoring module—and the augmentation and preprocessing strategies are designed to address the morphological richness and topic sparsity of Korean essay data described in Section 3. An overview of the entire proposed framework is illustrated in Figure 1, which summarizes the interaction among preprocessing, augmentation, and the Korean AES model architecture. To avoid redundancy, the architecture is described once below in mathematical form, with Figure 1 serving as the single visual reference.

[§]<https://github.com/monologg/KoBERT-Transformers>

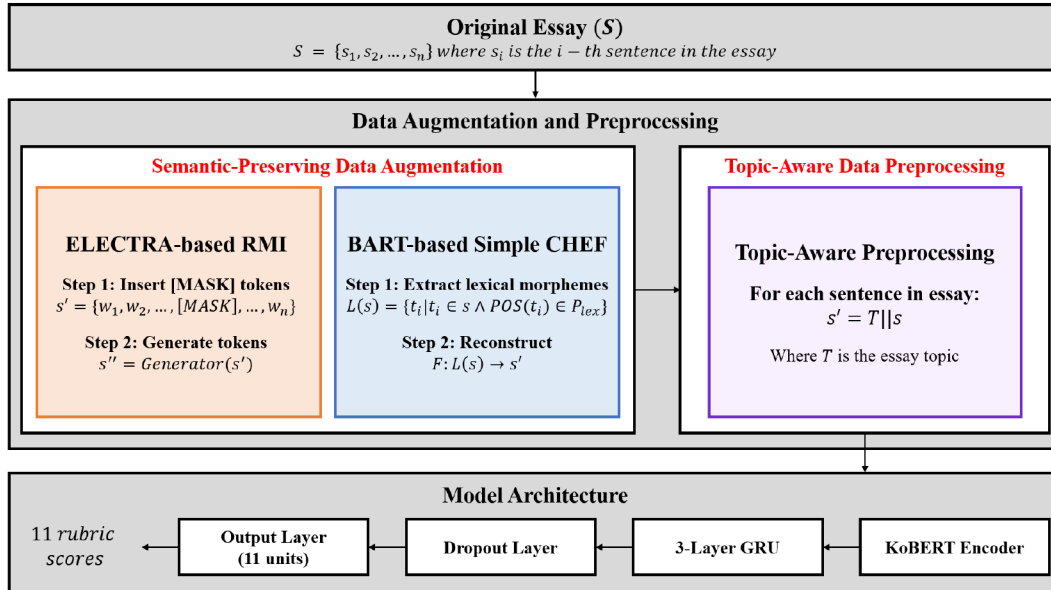


Figure 1. Proposed framework for improving the performance of Korean AES systems. The system consists of three core components: (1) linguistically informed data augmentation using CHEF and RMI to increase training diversity, (2) topic-aware data preprocessing that integrates prompt information into sentence inputs, and (3) a scoring model based on sentence-level embeddings extracted from a frozen KoBERT encoder and processed by a GRU-based prediction module.

4.1 Model Architecture

This proposed baseline model is built upon KoBERT, leveraging the *Essay Writing Evaluation Data* from *AI-HUB*. KoBERT is a transformer-based language model pretrained on a large-scale Korean corpus. Unlike multilingual BERT^h, KoBERT utilizes a Korean-specific vocabulary and morphological tokenization scheme, allowing for more linguistically appropriate representations of Korean text. In this model, KoBERT is used solely as a feature extractor for sentence-level embeddings and remains frozen during training. Each input essay is preprocessed into a sequence of sentences, denoted by $x_1, x_2, \dots, x_n$. Each sentence x_i is passed through the frozen KoBERT to obtain a contextualized embedding vector $\mathbf{e}_i \in \mathbb{R}^{768}$:

$$\mathbf{e}_i = \text{KoBERT}(x_i), \quad i = 1, 2, \dots, n \quad (2)$$

These sentence-level embeddings are padded or truncated at the front to a fixed sequence length of 128 sentences (so that the last position always holds the essay's final sentence) and then input into a three-layer GRU network. The GRU captures temporal dependencies across sentences and outputs the final essay-level representation $\mathbf{h}_{\text{final}} \in \mathbb{R}^{64}$:

$$\mathbf{h}_{\text{final}} = \text{GRU}_3([\mathbf{e}_1, \dots, \mathbf{e}_{128}]) \quad (3)$$

To prevent overfitting, a dropout layer with a rate of 0.5 is applied to the output:

$$\mathbf{h}_{\text{drop}} = \text{Dropout}(\mathbf{h}_{\text{final}}, p = 0.5) \quad (4)$$

The output layer is a fully connected dense layer with 11 units, each corresponding to one of the 11 rubric items defined in Section 3.2. Since all target scores are normalized to the $[0, 1]$ range by dividing the original 0–3 rubric scores by 3, a sigmoid activation is applied:

^h<https://huggingface.co/google-bert/bert-base-multilingual-cased>

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$$\hat{\mathbf{y}} = \sigma(\mathbf{W}_{\text{out}}\mathbf{h}_{\text{drop}} + \mathbf{b}_{\text{out}}) \quad (5)$$

Here, $\hat{\mathbf{y}} \in \mathbb{R}^{11}$ denotes the predicted normalized scores for each rubric. The model is optimized using the Mean Squared Error (MSE) loss averaged over all rubric items and across the batch:

$$\mathcal{L} = \frac{1}{11N} \sum_{j=1}^N \sum_{i=1}^{11} (y_{ij} - \hat{y}_{ij})^2 \quad (6)$$

where N is the batch size, y_{ij} is the normalized ground truth score, and $\hat{y}_{ij}$ is the predicted score for the i -th rubric item of the j -th essay.

4.2 Data Augmentation

This section introduces the data augmentation strategies employed in this study to address the limited availability of Korean essay data and enhance model generalization. Given the importance of preserving semantic integrity and lexical consistency in automated essay scoring, particular care was taken in selecting augmentation methods suitable for the linguistic characteristics of Korean. Specifically, two techniques—CHEF and RMI—were adopted, both of which are designed to minimize distortion while increasing data diversity in an efficient manner. Both methods are label-preserving: every augmented essay inherits the rubric scores of its source essay, and no additional human scoring is performed. Both methods also operate under pre-specified automatic filtering rules that discard degenerate generator outputs—for example, generations that are drastically truncated relative to the source sentence are rejected and the original sentence is retained. The exact criteria and thresholds are stated in Section 5.1. The semantic fidelity of both methods is verified quantitatively in Section 5.2 before being applied to the AES task. The following subsections provide detailed descriptions of each method.

4.2.1 BART-based CHEF: Morphology-Aware Augmentation

This study adopts the data augmentation framework proposed in [Seo et al. \(2023\)](#), termed **CHEF**, to generate linguistically valid sentence variants that reflect the *agglutinative structure of Korean*. CHEF leverages Korean's morphological richness, where lexical and grammatical morphemes are concatenated to form complex word units. For instance, the English sentence "*I eat an apple*" corresponds to the Korean sentence "*나는 사과를 먹는다*". While the English sentence comprises four independent words, its Korean counterpart is composed of three surface forms, each potentially containing multiple morphemes:

English: {I, eat, an, apple}
Korean: {나, 는, 사과, 를, 먹, 는다}

Here, 나 (I, pronoun), 사과 (apple, general noun), and 먹 (eat, verb) are *lexical morphemes* from our defined set $\mathcal{P}_{\text{lex}}$ that convey primary semantic content, whereas 는 (auxiliary particle), 를 (object particle), and 는다 (final ending) are *grammatical morphemes* that serve syntactic and pragmatic functions in Korean sentence construction. Korean's agglutinative structure enables stylistic diversity under semantic consistency by varying grammatical morphemes while fixing

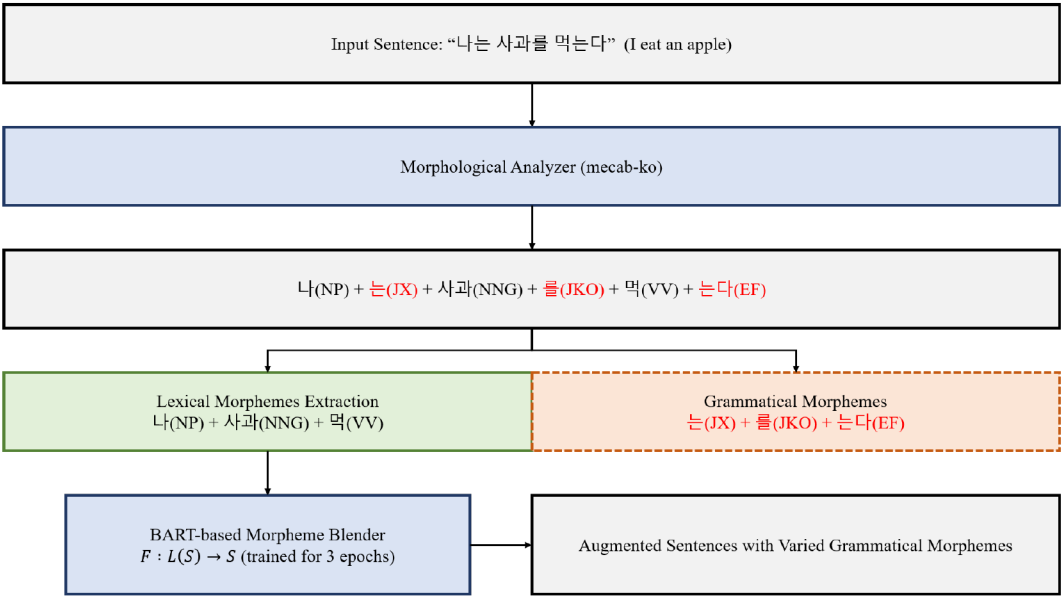


Figure 2. Structure of BART-based CHEF: Morphology-Aware Augmentation

lexical ones. Examples include:

- (1) plain/formal: 나는 사과를 먹는다
{나^{NP}, 는^{JX}, 사과^{NNG}, 를^{JKO}, 먹^{VV}, 는다^{EF}}
- (2) polite/formal: 나는 사과를 먹습니다
{나^{NP}, 는^{JX}, 사과^{NNG}, 를^{JKO}, 먹^{VV}, 습니다^{EF}}
- (3) polite/informal: 나는 사과를 먹어요
{나^{NP}, 는^{JX}, 사과^{NNG}, 를^{JKO}, 먹^{VV}, 어요^{EF}}

These expressions are semantically equivalent to "I eat an apple" in English, illustrating the generation capacity that CHEF exploits. The differences among them reflect stylistic variations commonly used in Korean, such as distinctions in formality and politeness, which are primarily encoded through sentence-final endings.

The CHEF framework comprises two main components: a morpheme blender and a label discriminator. The blender reconstructs the original sentence from lexical morphemes, while the discriminator filters syntactically invalid outputs. Following the simplified configuration in Seo et al. (2023), we exclude the discriminator and utilize only the morpheme extractor and blender, as shown in Figure 2. We utilize mecab-koⁱ as our morphological analyzer to assign part-of-speech (POS) tags to morphemes. For our implementation of CHEF, we define lexical morphemes based on a carefully selected subset of POS tags that capture the primary semantic content of Korean sentences, as detailed in Table 6. Our lexical morpheme set $\mathcal{P}_{\text{lex}}$ comprises 13 POS tags including nouns (NNG, NNP, NNB, NNBC), numerals and pronouns (NR, NP), verbs and adjectives (VV, VA), copulas and auxiliary verbs (VCP, VCN, VX), as well as numbers and determiners (SN, MM). This definition diverges from traditional linguistic categorizations by including copulas and auxiliary verbs, which are essential for Korean sentence construction and semantic completeness. All remaining

ⁱ<https://pypi.org/project/python-mecab-ko/>

Table 6. Definition of lexical morphemes in MeCab-ko POS tags for CHEF data augmentation.

Category	POS Tags	Description
Lexical Morphemes $\mathcal{P}_{\text{lex}}$	NNG, NNP, NNB, NNBC	General, proper, dependent, and unit nouns
	NR, NP	Numerals and pronouns
	VV, VA	Verbs and adjectives
	VCP, VCN, VX	Positive/negative copulas and auxiliary verbs
	SN, MM	Numbers and determiners
Grammatical Morphemes	JKS, JKC, JKG, JK0, JKB, JKV, JKQ	Case particles (subject, complement, genitive, object, adverbial, vocative, quotative)
	JC, JX	Conjunctive and auxiliary particles
	EP, EF, EC	Pre-final, final, and connective endings
	ETN, ETM	Nominalizing and adnominal endings
	XPN, XSN, XSV, XSA	Prefixes and derivational suffixes
	XR, MAG, MAJ, IC	Roots, adverbs, and interjections
Others	SF, SE, SS0, SSC, SC	Punctuation marks and delimiters
	SY, SH, SL	Other symbols, Chinese characters, foreign words

morphemes, including case particles, endings, and various functional elements, are classified as grammatical morphemes and are excluded from the input representation.

For generation, we employ mBART-50^j, which has demonstrated superior performance in Korean generation tasks compared to KoBART^k. According to evaluations on the KommonGen benchmark (Seo et al. 2022)—a commonsense-aware generation dataset for assessing grammaticality, fluency, and conceptual coverage—mBART-50 consistently outperforms KoBART and KoGPT2, demonstrating strong generalization ability in Korean sentence generation tasks.

Formally, given an original sentence S , the lexical morpheme set $\mathcal{L}(S)$ is defined as:

$$\mathcal{L}(S) = \{t_i \mid t_i \in S \wedge \text{POS}(t_i) \in \mathcal{P}_{\text{lex}}\}, \quad (7)$$

where $\mathcal{P}_{\text{lex}} = \{\text{NNG, NNP, NNB, NNBC, NR, NP, VV, VA, VCP, VCN, VX, SN, MM}\}$ denotes the complete set of POS tags for lexical morphemes as defined in our study, and t_i represents a morpheme token in the sentence. The morpheme blender learns a mapping function $\mathcal{F} : \mathcal{L}(S) \rightarrow S$ to reconstruct coherent and grammatically valid sentences from the lexical morpheme input. The model is trained for three epochs on parallel sentence pairs $(\mathcal{L}(S), S)$, where the generated output $\hat{S} = \mathcal{F}(\mathcal{L}(S))$ serves as augmented training data. This approach leverages Korean's agglutinative morphology to generate stylistically diverse variants while preserving semantic content, thereby enhancing the robustness and generalization capability of downstream models. Because the blender receives the full set of lexical morphemes of the source sentence, the content vocabulary of an augmented sentence stays close to what the student actually wrote—a useful property for keeping augmented data appropriate to the school level of the source essay. Two qualifications are needed, however. First, this is a property of the input representation, not a hard constraint on the output: the blender is a generative model and may still introduce a word of its own. Second, and more importantly, our lexical set $\mathcal{P}_{\text{lex}}$ follows the operational definition of Seo et al. (2023) and therefore excludes adverbs (MAG, MAJ), roots (XR) and interjections (IC), which are content-bearing words in the linguistic sense; these are withheld from the blender's input and must be regenerated. The resulting departures are measured over the full corpus in Section 5.2 rather than assumed.

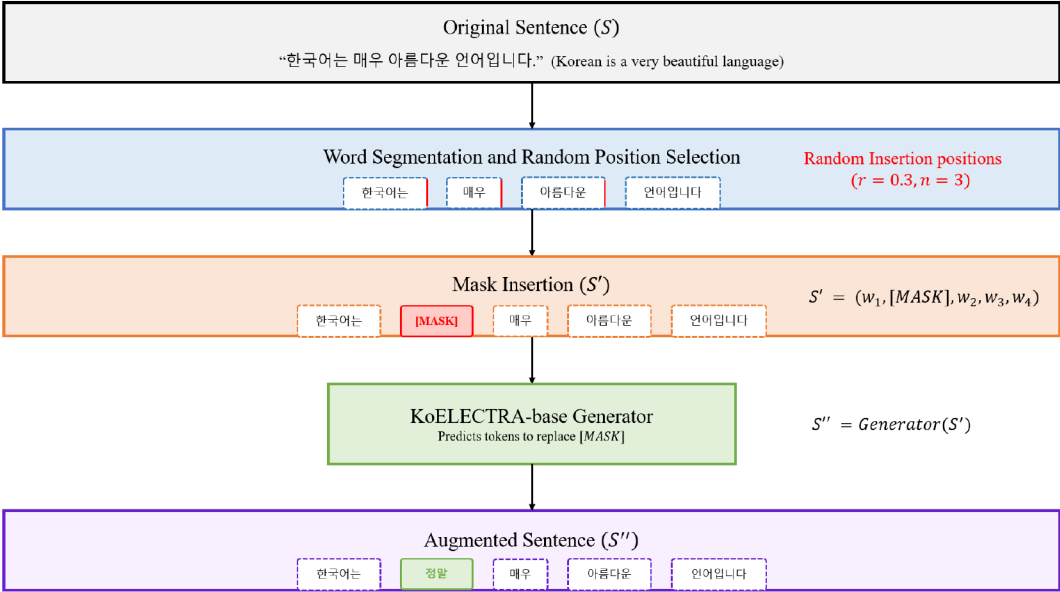


Figure 3. Structure of ELECTRA-based RMI: Random Masking Insertion

4.2.2 ELECTRA-based RMI: Random Masking Insertion

In addition to CHEF, we adopt Random Masking Insertion (RMI) as an auxiliary data augmentation technique to further improve model robustness and generalization in the Korean AES task. RMI builds upon the foundational ideas introduced in Yang (2022), while incorporating several key modifications to better accommodate the morphological and syntactic characteristics of the Korean language. A widely used predecessor of RMI is Random Masking Replacement (RMR), which originated from synonym substitution techniques and was later formalized in the Easy Data Augmentation (EDA) framework (Wei and Zou 2019). Early synonym replacement methods employed static lexical resources or dictionaries, while more recent approaches have utilized masked language models (MLMs) to generate context-sensitive replacements (Kumar et al. 2020; Kim et al. 2020; Wu et al. 2019; Kobayashi 2018). These methods have been shown to generate more natural and semantically coherent outputs, particularly in English. However, applying RMR directly to Korean often results in syntactically awkward or semantically incoherent outputs, primarily due to the agglutinative nature of the language (Seo and Oh 2023; Yang 2022). In Korean, words are composed of a root lexical morpheme followed by one or more grammatical morphemes (e.g., case particles, verb endings), and these must agree morphologically and syntactically with their surrounding elements. As emphasized in Seo et al. (2023), the morphological validity of particle forms is highly dependent on their associated stems. Consequently, naïve token replacement without morphological realignment frequently produces ungrammatical results. Furthermore, Korean exhibits significant lexical variability due to factors such as Sino-Korean versus native vocabulary, loanword integration, polysemy, and non-standardized spacing conventions. Combined with the presence of over nine distinct particle types, this variability makes fine-tuned replacement strategies both computationally expensive and linguistically fragile. To address these challenges, we adopt a modified approach that performs insertion-based augmentation rather than replacement. Specifically, RMI introduces masked tokens at random positions in

^j<https://huggingface.co/facebook/mbart-large-50>

^k<https://github.com/SKT-AI/KoBART>

the sentence and uses a pretrained masked language model to predict the appropriate insertions. This strategy is particularly well-suited to Korean, where flexible constituent order permits the insertion of words at various positions without severely affecting grammaticality or meaning (Kim et al. 2024). An important feature of RMI is that it preserves the original lexical morphemes in the sentence, thereby minimizing semantic drift—a crucial property for AES tasks, which demand high content fidelity. Moreover, insertion, as opposed to substitution, avoids corrupting morphologically bound expressions and yields more structurally coherent outputs. For implementation, we employ the KoELECTRA¹-base generator, which has been pretrained on 34GB of Korean text. Insertions are made based on a fixed ratio parameter r , which determines the proportion of inserted tokens relative to the sentence length. To preserve morphological coherence, insertions are performed at word-level boundaries based on whitespace splitting, rather than at token-level positions.

Formally, given an input sentence S , represented as a sequence of words:

$$S = (w_1, w_2, \dots, w_n), \quad (8)$$

we sample insertion positions uniformly at random and insert [MASK] tokens, resulting in a masked sentence S' :

$$S' = (w_1, w_2, \dots, [\text{MASK}], \dots, w_n). \quad (9)$$

The modified sentence S' is then passed through the KoELECTRA generator to predict replacements for the inserted tokens:

$$S'' = \text{Generator}(S'), \quad (10)$$

where S'' denotes the final augmented sentence. The insertion process is repeated $\lfloor n \cdot r \rfloor$ times per sentence, where $n = \text{len}(S) - 1$, producing multiple diverse variants from a single original sentence. In all AES experiments in this study, the insertion ratio is fixed at $r = 0.3$.

As illustrated in Figure 3, each instance of RMI augmentation maintains the structural integrity of the original sentence while introducing controlled lexical variation. This approach effectively increases training diversity and helps the AES model become more robust to stylistic and lexical variation commonly observed in Korean learner essays.

4.3 Topic-Aware Data Preprocessing

Essay scoring is inherently influenced by the relevance, coherence, and alignment of the essay content to the given prompt or topic. As scoring rubrics in AES systems often include topical relevance as an explicit criterion, enabling the model to account for prompt-specific information can significantly enhance its ability to evaluate essays more holistically and accurately. Prompt- and topic-aware scoring is an established direction in AES: Gupta (2023) demonstrated that integrating prompt information into model inputs substantially improves AES performance, Cho et al. (2021) applied topic-aware scoring to Korean AES, and Jiang et al. (2023) studied prompt-aware scoring from a domain-generalization perspective. Following the findings of Gupta (2023), we implement a topic-aware preprocessing strategy designed to inject topic context at the sentence level. This design choice aligns with our architecture, wherein KoBERT operates on sentence-level inputs and independently encodes each sentence into contextual embeddings (as illustrated in Figure 1). Given that KoBERT does not natively model inter-sentence dependencies, encoding the prompt at the sentence level ensures that topic relevance is available to the model during each inference step.

¹<https://huggingface.co/monologg/koelectra-base-v3-generator>

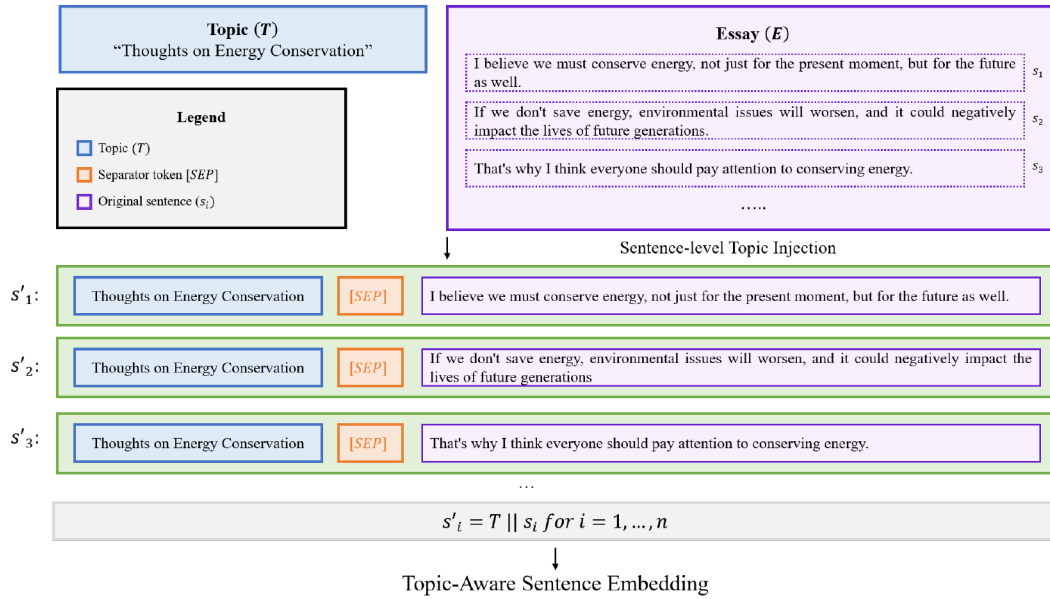


Figure 4. Overall process of the topic-aware data preprocessing

In the *Essay Writing Evaluation Data*, each essay E is associated with a corresponding topic label T . The essay is first segmented into a sequence of n sentences:

$$E = (s_1, s_2, \dots, s_n), \quad (11)$$

where s_i denotes the i -th sentence of essay E . Each sentence is then transformed into a topic-aware input s'_i by prepending the topic label and a separator token:

$$s'_i = T \parallel [\text{SEP}] \parallel s_i, \quad \text{for } i = 1, \dots, n, \quad (12)$$

where $\parallel$ denotes string concatenation and $[\text{SEP}]$ is a special token recognized by KoBERT as a segment boundary. This preprocessing step ensures that topic information is explicitly visible to KoBERT at the input layer, enabling the model to construct contextual embeddings that encode both the semantic content of the sentence and its topical alignment with the given prompt. This strategy offers several advantages. First, it avoids altering the internal structure or syntax of the original sentence, preserving linguistic integrity. Second, it provides lightweight but consistent supervision by embedding the same topic representation across all sentences in the essay. Finally, by aligning input representations with the scoring rubric's topical relevance criterion, this method enables the scoring model to better capture semantic relevance at a fine-grained level. As illustrated in Figure 4, the topic labeling process is deterministic and requires no additional annotation, making it suitable for efficient integration in training pipelines. Our cross-validation is stratified by prompt, so all 50 prompts occur in both the training and the validation portion of every fold; the gains reported in Section 5.4 should therefore be read as gains under prompt-matched conditions. Through this approach, we hypothesize that the model can better distinguish between on-topic and off-topic responses—an essential capability in both formative feedback and high-stakes writing assessment scenarios.

5. Experiments

This section presents a series of experiments designed to evaluate the effectiveness of the proposed methods for enhancing Korean AES. Specifically, we investigate two main components: (1) linguistically informed data augmentation, and (2) topic-aware preprocessing.

The experimental evaluation comprises four primary aspects:

1. **Semantic Preservation:** Assessing whether augmented data generated by CHEF and RMI methods preserve the semantic integrity of the original texts.
2. **Baseline Comparison:** Situating the proposed baseline relative to the official reference models distributed with the corpus and traditional machine learning methods under an identical protocol, before evaluating the proposed methods on top of it.
3. **Model Performance:** Evaluating the impact of the proposed augmentation techniques and topic-aware preprocessing on AES accuracy, with formal statistical significance testing for every comparison and an analysis of sensitivity to the augmentation ratio.
4. **Combined Effects:** Formally testing whether augmentation provides additional gains on top of topic-aware preprocessing.

5.1 Experimental Settings and Evaluation Metric

All experiments were conducted under consistent settings. The AES model was trained on an NVIDIA RTX 3090 Ti GPU. Each sentence is encoded independently by the frozen KoBERT with a maximum of 100 word-piece tokens, and the final-layer [CLS] vector is taken as the 768-dimensional sentence representation. An essay is thus a sequence of sentence vectors; sequences are truncated or zero-padded to 128 sentences, in both cases at the *front* (“pre” mode), so that the last position always holds the final sentence of the essay. The scoring module consists of a 3-layer GRU (hidden sizes $256 \rightarrow 128 \rightarrow 64$) with dropout 0.4 applied to the input of each layer; the essay representation is the final hidden state of the third layer, which the pre-padding convention makes the state after the last real sentence. It passes through dropout 0.5 into a dense output layer with 11 sigmoid units. Training uses Adam with a learning rate of 5×10^{-4} , decayed by $\exp(0.1(10 - e))$ after epoch $e = 10$, a batch size of 64, MSE loss, at most 50 epochs, and a fixed random seed of 42.

Augmentation generation settings. The augmented sets are produced once, with a fixed seed of 42, under the following criteria. For CHEF, the mBART-50 morpheme blender is fine-tuned for three epochs as described in Section 4.2, and a length-preservation rule is applied to its output: a blended sentence whose word count falls below 80% of its source sentence is discarded and the source sentence is retained in its place, so that a degenerate, heavily truncated blend can never replace the original. For RMI, the insertion ratio is $r = 0.30$, giving $\lfloor n \cdot r \rfloor$ inserted masks per sentence as defined in Equation 10; each mask is filled with a token sampled from the ten highest-scoring predictions of the KoELECTRA generator, and sentences of fewer than two words are returned unchanged. These are the pre-specified filtering rules referred to in Section 4.2; they are applied uniformly to every generated sentence, not to a sampled subset.

For performance evaluation, we employ **10-fold cross-validation stratified by prompt**, so that all 50 prompts are proportionally represented in every fold. We choose this protocol for two reasons: a 10-fold split maximizes the training data available per fold while still yielding enough folds for stable estimates and paired significance testing, and stratification by prompt removes prompt-distribution imbalance across folds—an important consideration given that each of the 50 prompts contributes only about 900 essays. All reported values are means over the 10 folds accompanied by the fold-level sample standard deviation (computed with $n - 1$ in the denominator).

Statistical significance protocol. For every comparison between a method and its reference configuration, we conduct a paired one-sided t -test over the 10 per-fold differences (method – reference; H_1 : method > reference, $n = 10$, $\alpha = 0.05$); potential degradations are tested two-sided. The one-sided alternative reflects the directional hypothesis fixed before experimentation—each component is introduced specifically to improve performance—and the corresponding two-sided p -values are twice those reported. The FCS-level comparisons in Table 10 constitute the confirmatory analyses of this study; the per-rubric tests are reported as exploratory and are not corrected. Because the confirmatory family contains more than one test, we report for each comparison, alongside the raw p -value, the Holm–Bonferroni adjusted p -value over that family and the p -value of a paired one-sided Wilcoxon signed-rank test, which does not assume normality of the per-fold differences. A claim is called significant in this paper only when it survives the Holm adjustment; comparisons that are nominally but not adjusted-significant are reported as such and described as suggestive.

Three caveats apply to the testing procedure itself. Because cross-validation folds share most of their training data, fold-level paired t -tests are known to be potentially anti-conservative (Dietterich 1998; Nadeau and Bengio 2003). All experiments use a single random seed (42), so seed-level variance is not captured; with effect sizes of $\Delta \approx 0.005$ against a fold-level standard deviation of ≈ 0.012 , replication across seeds would be the natural strengthening of this evidence, and we did not perform it. Finally, the ten fold-level differences are not independent observations of a population, so the p -values quantify the consistency of an effect across a fixed partition of one corpus rather than generalization to new corpora. Taken together, these caveats mean that the correct reading of a nominally significant borderline result here is “consistent in direction across folds,” not “established.” Throughout this paper, the word “significant” is used only for differences confirmed by these tests. In all tables, * and ** denote $p < 0.05$ and $p < 0.01$ for improvements, and † denotes a statistically significant decrease. We report four decimal places because the effect sizes at issue ($\Delta \approx 0.004$ – 0.008) would be invisible at two decimal places.

Leakage control and two disclosures. Augmented data are added only to the training folds. Validation folds always consist exclusively of original essays, so no augmented text or its label is ever used for evaluation. Identical fold splits and settings are used for every configuration being compared. Two properties of the protocol should nevertheless be stated explicitly. First, training stops early when the loss on the validation fold has not improved for 5 epochs, and the parameters of the best such epoch are restored; the fold used for model selection is therefore also the fold that is scored. We retain this procedure because it is the one used by the reference implementation for this corpus, and because it is applied identically to every configuration, so it cannot favour one over another—but it does mean that the absolute QWK values carry a mild optimistic bias, and they should be read as internally comparable rather than as unbiased held-out estimates. Second, the two generators used for augmentation, as well as the word2vec embeddings of Section 5.3, are fitted once on the text of the whole corpus rather than separately within each training fold. This exposure is label-free—the CHEF blender is trained only to reconstruct a sentence from its own lexical morphemes, and word2vec only sees raw token sequences; no rubric score is visible to either—and augmented essays never enter a validation fold. Per-fold refitting of the generators would remove even this indirect exposure, at ten times the generation cost, and we note it as a deviation from the strictest protocol.

As the main evaluation metric, we used Quadratic Weighted Kappa (QWK), commonly adopted in ordinal classification tasks such as AES. Given that each of the 11 rubric scores ranges from 0 to 3, QWK was computed individually for each item. We additionally calculated QWK on the FCS, defined in Equation 1, after quartile-based discretization into four classes as described in Section 3.2.

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}} \quad (13)$$

where $O_{i,j}$ is the observed agreement matrix, $E_{i,j}$ the expected agreement matrix, and $w_{i,j} = \frac{(i-j)^2}{(k-1)^2}$ is the quadratic weight penalizing larger class disagreements.

5.2 Semantic Preservation and Augmentation Quality Validation

Before we directly evaluate the impact of augmented data on AES performance, we conduct a separate task-specific evaluation to assess the semantic fidelity of each augmentation method. Rather than manually inspecting a small number of examples, our validation strategy combines (i) pre-specified automatic filtering rules applied uniformly to the entire generation pipeline, which discard degenerate generator outputs before any augmented sentence enters the training data, and (ii) the quantitative, corpus-wide evaluations described below. This ensures that every augmented sentence—not merely a sampled subset—passes the same validity checks. We did not perform exhaustive manual verification of the augmented data, which we acknowledge as a limitation; however, the two quantitative evaluations below directly measure the properties that manual inspection would target (grammatical validity and meaning preservation) at the scale of the full dataset.

To this end, we utilize the KLUE-STs dataset, a standard benchmark for sentence-level semantic similarity in Korean. KLUE-STs is part of the Korean Language Understanding Evaluation (KLUE) benchmark (Park et al. 2021), which is designed to assess various core NLP capabilities across multiple tasks. The STs (Semantic Textual Similarity) component specifically measures how well models capture the degree of semantic equivalence between two given sentences on a continuous scale. We first train a baseline STs model using the original KLUE-STs training set and measure its performance on the official test set. We then apply each augmentation technique—CHEF, RMI, EDA, and RMR—to expand the training data, retrain the model on each augmented set, and compare the resulting STs performance. Since the STs task requires fine-grained semantic matching between sentence pairs, any degradation in model accuracy suggests that the augmented data may have introduced semantic drift. Conversely, maintained or improved performance indicates better semantic preservation. In addition to task-level evaluation, we also compute cosine similarity between sentence embeddings of original–augmented pairs across the KLUE-STs training set. This metric provides a direct, model-agnostic estimate of semantic closeness. Taken together, these two evaluations—STs performance and cosine similarity—offer both functional and structural perspectives on how well each augmentation method preserves meaning. This dual assessment serves as a prerequisite to ensure the semantic reliability of augmented samples before applying them to the AES task.

With respect to school-level appropriateness—whether augmented sentences remain plausible outputs for students of the source essay’s grade—the two methods differ, and we quantify the difference rather than assert it. For every original/augmented sentence pair in the corpus we tagged both sides with mecab-ko and counted morphemes present in the augmented sentence but not in the source, grouped into three classes: the 13 tags of $\mathcal{P}_{\text{lex}}$; adverbs, roots and interjections (MAG, MAJ, XR, IC), which are content words that our lexical set does not cover; and function morphemes.

For CHEF (512,544 sentence pairs), 13.5% of augmented sentences contain at least one morpheme from $\mathcal{P}_{\text{lex}}$ that the source did not have, and 15.1% contain a new adverb, root or interjection, at an average of 0.17 new content morphemes per sentence; the modifications are overwhelmingly grammatical, with connective endings (EC, 17.5%), auxiliary particles (JX, 9.7%) and case particles making up the bulk of the changed material. CHEF therefore keeps content vocabulary close to the source but does not guarantee it: roughly one sentence in seven acquires

Table 7. Quantitative evaluation of semantic preservation across augmentation methods, measured via cosine similarity and Pearson correlation on the KLUE-STS benchmark.

Augmentation Method	Cosine Similarity	STS (Pearson’s r)
Original (no augmentation)	–	0.91986
Easy Data Augmentation (EDA)	0.94629	0.92016
Random Masking Replacement (RMR)	0.93504	0.89762
Random Masking Insertion (RMI)	0.97817	0.92353
CHEF	0.95010	0.92753

a content word the student did not write. For RMI (515,565 pairs) the picture is necessarily different, since the method inserts new material by construction: 73.9% of sentences receive at least one new morpheme from $\mathcal{P}_{\text{lex}}$, and the inserted material is 32.2% adverbs and 6.2% conjunctive adverbs, but also 17.5% common nouns and 9.8% verbs. Insertions are thus not, as one might assume from the examples in Table 8, restricted to function words and discourse adverbs: about a third of them are nouns, verbs or adjectives. Because the KoELECTRA generator is pretrained on general-domain adult text, this leaves a real possibility that an inserted content word exceeds the vocabulary level of the source grade. We did not conduct a human evaluation of grade-level plausibility, which remains the appropriate way to settle the question and which we leave as a limitation; the measurements above delimit how large the issue can be, and indicate that it is small for CHEF and non-negligible for RMI. A minor related note: CHEF generation succeeded for 44,383 of the 44,658 essays (99.4%), the remainder being dropped by the filtering rules, so the CHEF “1:1” condition adds 0.994 augmented essays per original.

A related consideration concerns the label-preserving assumption for rubric items that assess surface form rather than meaning—Grammar (exp1), Sentence (exp3), and Length (org2). Semantic preservation alone does not guarantee that inherited labels remain exact for such items; for instance, RMI insertions slightly increase word counts. Two factors mitigate this concern. First, augmented essays are used exclusively for training, and every evaluation is conducted on original essays only (Section 5.1), so any label imprecision on augmented samples acts as bounded training noise rather than evaluation bias. Second, the empirical results provide a direct check: none of the form-related items is significantly degraded in any of the configurations reported in Tables 11–13, and Length (org2) in fact improves significantly under both augmentation methods (Table 11), indicating that the inherited labels remain adequate as training targets. Finally, we note that the quantitative preservation evaluations below are conducted on KLUE-STS sentences rather than on the essay corpus itself; transferring these estimates to essay data is an assumption that we make explicit here.

5.2.1 Quantitative Evaluation

Building on the semantic preservation framework described above, we perform a quantitative evaluation to empirically compare the semantic fidelity of each augmentation method. Specifically, we assess semantic consistency using two complementary metrics introduced earlier: cosine similarity between original–augmented sentence embeddings and Pearson correlation on the KLUE-STS benchmark. Cosine similarity captures embedding-level proximity and reflects lexical and contextual closeness, while STS performance provides a task-based measure of how well augmented data supports semantic similarity prediction. Together, these metrics offer a comprehensive view of both structural and functional semantic alignment. Table 7 summarizes the results. For cosine similarity, we compute the average cosine score between sentence embeddings of original and augmented texts using the multilingual MiniLM model^m. For STS evaluation, we retrain a semantic similarity model on KLUE-STS using data augmented by each method and report Pearson’s r

^m<https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>

Table 8. Representative examples of data augmentation using CHEF and RMI. Bold tokens indicate modified elements, with CHEF preserving lexical morphemes while varying grammatical morphemes.

Original Sentence	Augmented (CHEF)	Augmented (RMI)
대한민국 사람들은 이 날이 되면 팥죽을 먹는다. (People in South Korea eat red bean porridge on this day.)	대한민국 사람들은 이 날이 되면 팥죽을 먹습니다 . (EF variation, plain → formal)	대한민국 사람들은 보통 이 날이 되면 팥죽을 먹는다. (Adverb insertion)
그 후 강아지를 입양하여 키우기 시작하였습니다. (After that, I adopted a puppy and started raising it.)	그 후 강아지를 입양해 키우기 시작했습니다 . (EC variation, contraction)	그 후 에 강아지를 입양하여 키우기 시작하였습니다. (Postposition insertion)
그렇기에 경찰이 존경을 받는 것입니다. (That's why the police are respected.)	그렇기에 경찰들은 존경을 받는 것입니다. (JKS → JI variation)	그렇기에 경찰이 존경을 많이 받는 것입니다. (Adverb insertion)

on the test set. All augmentation methods are applied with a 1:1 expansion ratio. The results reveal notable differences in semantic preservation across techniques. Random Masking Insertion (RMI) achieves the highest cosine similarity score (0.97817), indicating minimal lexical and contextual distortion at the embedding level. This suggests that RMI alters the sentence surface minimally, thereby preserving latent semantics. However, its STS performance is slightly lower than CHEF, implying that lexical-level fidelity does not always translate to functional semantic equivalence in human-judged similarity tasks. CHEF, by contrast, yields the highest STS Pearson correlation (0.92753), reflecting its strength in preserving core sentence meaning from a human-aligned perspective. This result supports the intuition that CHEF's morphosyntactic rewritings—such as varying particles and verb endings while keeping lexical morphemes intact—strike a better balance between syntactic variation and semantic retention. Random Masking Replacement (RMR) performs worse on both metrics, indicating a greater degree of semantic drift. Its token-level substitutions frequently lead to ungrammatical or semantically incoherent sentences, as corroborated by its lower cosine similarity (0.93504) and STS correlation (0.89762). Easy Data Augmentation (EDA) shows moderate performance on both metrics but exhibits less consistency and linguistic control. While it achieves a reasonable cosine similarity score (0.94629) and STS correlation (0.92016), these results likely stem from occasional surface-level edits rather than systematic semantic preservation. Due to its reliance on random perturbations such as synonym replacement, word deletion, and shuffling—without any linguistic supervision—EDA often introduces subtle grammatical errors or stylistic inconsistencies. This makes it less suitable for high-stakes tasks like AES, where preserving both meaning and syntactic integrity is critical. Taken together, these results validate our design choice to favor linguistically informed augmentation strategies—CHEF and RMI—for the AES task. CHEF is more effective at preserving human-perceived meaning, while RMI excels in embedding-level consistency. Their complementary strengths make them well-suited for generating diverse yet semantically reliable training data.

5.2.2 Qualitative Examples

To complement the quantitative results, Table 8 provides illustrative examples of augmented sentences produced by CHEF and RMI, demonstrating how each method alters surface forms while maintaining the original meaning. These examples highlight the linguistic operations underlying each augmentation strategy and offer qualitative insight into their semantic reliability.

In the first example, “대한민국 사람들은 이 날이 되면 팥죽을 먹는다” (Daehanminguk saramdeureun i nari doemyeon patjuk-eul meokneunda; “People in South Korea eat red bean porridge on this day”), CHEF replaces the plain declarative ending 먹는다 (meokneunda) with the formal polite 먹습니다 (meokseumnida), which reflects a morphosyntactic variation in the

Table 9. Comparison of the proposed baseline with re-implementations of the official corpus reference models and with traditional machine learning methods (FCS QWK; identical 10-fold prompt-stratified CV and splits). Δ = alternative – KoBERT+GRU; p from a paired one-sided t -test (H_1 : KoBERT+GRU > alternative, $n = 10$).

Family	Model	Features	FCS QWK	Δ	p (one-sided)
Proposed baseline	KoBERT + GRU	frozen KoBERT 768d	0.4243 $\pm$ 0.0120	—	—
Official corpus baselines	word2vec + LSTM	corpus-trained w2v 200d	0.3861 $\pm$ 0.0143	−0.0382	1.4 $\times 10^{-7}$
	word2vec + GRU	corpus-trained w2v 200d	0.3740 $\pm$ 0.0133	−0.0503	4.0 $\times 10^{-9}$
Traditional ML	kNN ($k=20$)	KoBERT sent-mean 768d	0.3732 $\pm$ 0.0144	−0.0511	1.0 $\times 10^{-6}$
	Random Forest (300 trees)	KoBERT sent-mean 768d	0.3195 $\pm$ 0.0121	−0.1048	9.7 $\times 10^{-10}$
	Ridge regression	KoBERT sent-mean 768d	0.3015 $\pm$ 0.0098	−0.1228	4.4 $\times 10^{-11}$
	Linear SVR	KoBERT sent-mean 768d	0.3004 $\pm$ 0.0095	−0.1239	7.2 $\times 10^{-11}$
	Decision Tree	KoBERT sent-mean 768d	0.2423 $\pm$ 0.0121	−0.1820	4.3 $\times 10^{-12}$

sentence-final ending (EF). This transformation adjusts the formality of the sentence without altering its meaning. In contrast, RMI inserts the adverb 보통 (botong; “usually”), introducing a habitual nuance. The insertion maintains the original propositional content while enhancing contextual richness.

In the second example, “그 후 강아지를 입양하여 키우기 시작하였습니다” (Geu hu gangajireul ipyanghayeo kiu-gi sijakhaetsseumnida; “After that, I adopted a puppy and started raising it”), CHEF applies two morphological changes: first, it contracts the connective verb form 입양하여 (ipyanghayeo) into the more colloquial 입양해 (ipyanghae), and second, it shifts the verb ending from 시작하였습니다 (sijakhaetsseumnida) to 시작했습니다 (sijakhaetseumnida). Both transformations involve changes to grammatical morphemes while preserving core lexical items and meaning. RMI inserts the postposition 에 (e), modifying the temporal phrase from 그 후 (geu hu; “after that”) to 그 후에 (geu hue), which is more naturally aligned with spoken Korean but semantically redundant.

In the third example, “그렇기에 경찰이 존경을 받는 것입니다” (Geureotgie gyeongchali jongyeonggeul batneun geosimnida; “That’s why the police are respected”), CHEF modifies the subject noun phrase 경찰이 (gyeongchali) to 경찰들은 (gyeongchal-deureun), changing the subject particle (JKS) to a topic particle (JX) and introducing plural marking (-들). This shift subtly broadens referential scope and discourse emphasis while maintaining meaning. RMI adds the adverb 많이 (mani; “a lot”), intensifying the predicate 받는 (batneun; “receiving”). The modification reinforces the degree of respect without introducing semantic drift.

Together with the quantitative results, the qualitative examples further highlight the complementary nature of the two proposed augmentation methods. CHEF systematically applies morphosyntactic rewritings—such as variations in honorific forms, connective endings, and case particles—while preserving core lexical elements and enhancing stylistic diversity. In contrast, RMI introduces minimal lexical insertions (e.g., adverbs and postpositions) that retain sentence structure and meaning, offering subtle surface-level variation. These linguistically grounded strategies provide distinct but synergistic approaches to data augmentation, supporting the creation of semantically faithful yet diverse training samples that contribute to more robust and generalizable AES models.

5.3 Baseline Comparison with Reference Models and Traditional Machine Learning

As a supplementary experiment, we situate the KoBERT-based baseline relative to other model families on this corpus. This comparison serves two purposes: it examines whether more traditional machine learning methods are competitive at this data scale, and it relates the baseline to the reference models previously reported for this corpus, under a controlled protocol (same 10-fold prompt-stratified splits, same seed, same targets; Table 9).

Table 10. Summary of the five main comparisons on the FCS QWK (10-fold prompt-stratified CV, mean $\pm$ std). Δ = method – reference; p from a paired one-sided t -test over per-fold differences ($n = 10$); “folds won” counts folds where the method outperformed its reference. “+Topic” denotes the topic-aware preprocessing of Section 4.3; the sentence encoder is identical in all rows, so every comparison isolates a single component. The Holm column adjusts over these five comparisons, which form the confirmatory family; bold marks the three that survive the adjustment.

#	Comparison	Reference QWK	Method QWK	Δ	p (one-sided)	p (Holm)	p (Wilcoxon)	Folds won
1	CHEF 1:1 vs. Baseline	0.4243 $\pm$ 0.0120	0.4318 $\pm$ 0.0115	+0.0075	0.0044	0.0177	0.0049	8 / 10
2	RMI ($r=0.3$) 1:1 vs. Baseline	0.4243 $\pm$ 0.0120	0.4291 $\pm$ 0.0133	+0.0048	0.0106	0.0319	0.0098	8 / 10
3	+Topic vs. Baseline	0.4243 $\pm$ 0.0120	0.4311 $\pm$ 0.0124	+0.0068	0.0008	0.0041	0.0049	9 / 10
4	CHEF 1:1 + Topic vs. +Topic	0.4311 $\pm$ 0.0124	0.4360 $\pm$ 0.0122	+0.0048	0.0448	0.0895	0.0654	7 / 10
5	RMI ($r=0.3$) + Topic vs. +Topic	0.4311 $\pm$ 0.0124	0.4347 $\pm$ 0.0134	+0.0035	0.0464	0.0895	0.0801	7 / 10

First, we re-implemented the official reference models distributed with the corpus—word2vec-based LSTM and GRU scorers—using our preprocessing and evaluation pipeline.ⁿ Both score below the KoBERT+GRU baseline (-0.0382 and -0.0503 FCS QWK, respectively) in all 10 folds ($p < 10^{-6}$). Within the word2vec family, upgrading the recurrent cell from GRU to LSTM helps ($+0.0121$, $p = 0.0004$), suggesting that the representation—rather than the recurrent architecture—is the dominant factor in the remaining gap.

Second, five traditional machine learning methods (kNN, Random Forest, Ridge, Linear SVR, and Decision Tree), each given the same frozen-KoBERT sentence embeddings mean-pooled per essay, also score below the baseline in all 10 folds, with differences ranging from -0.0511 (kNN) to -0.1820 (Decision Tree), all $p < 10^{-5}$.^o Taken together, these comparisons establish concrete reference points for the results that follow and indicate that, despite the modest corpus size, models built on pretrained representations remain the more effective choice on this data. All subsequent experiments therefore build on the KoBERT+GRU baseline.

5.4 Main Results

Table 10 summarizes the five central comparisons of this study on the holistic FCS metric. The per-rubric results are reported in three tables, one per experimental question, so that every significance marker is read against a single, unambiguous reference configuration: the effect of augmentation alone (Table 11), the effect of topic-aware preprocessing (Table 12), and the effect of augmentation on top of topic-aware preprocessing (Table 13). Three findings emerge.

Both augmentation methods yield significant improvements. Relative to the baseline (FCS QWK 0.4243 $\pm$ 0.0120), CHEF at a 1:1 ratio improves the FCS QWK to 0.4318 ($\Delta = +0.0075$, $p = 0.0044$, 8/10 folds) and RMI with $r = 0.3$ at a 1:1 ratio to 0.4291 ($\Delta = +0.0048$, $p = 0.0106$, 8/10 folds). Average rubric QWK also improves significantly for both (CHEF $p = 0.0197$; RMI $p = 0.0002$). At the rubric level, the largest and most robust gains concentrate on *Novelty* (con2), the weakest item of the baseline: CHEF raises its QWK from 0.1428 to 0.1786 ($+0.0358$, $p = 0.0010$) and RMI to 0.1742 ($+0.0314$, $p = 0.0020$), while near-ceiling items (≈ 0.95) are preserved (Table 11). Importantly, no rubric item is significantly degraded by either method.

Topic-aware preprocessing contributes independently. Adding topic-aware preprocessing alone raises the FCS QWK to 0.4311 ($\Delta = +0.0068$, $p = 0.0008$, 9/10 folds) and the average rubric QWK to 0.5768 ($p < 0.0001$), with broad, highly significant per-item gains on Vocabulary, Sentence, Length, In-paragraph, Structure, and Prompt (Table 12). Injecting the prompt into the sentence-level input therefore supplies a signal that augmentation alone does not. The breadth

ⁿOur re-implementation uses MeCab tokenization with 200-dimensional word2vec embeddings trained on the corpus, which differs from the distributors’ original embedding setup; it should therefore be read as a faithful representational reproduction rather than a bit-exact one.

^oBecause $n \gg p$ here (about 40,000 essays against 768 features), Linear SVR is fitted with the primal solver and the squared ϵ -insensitive loss; it converges in every fold under a budget of 20,000 iterations. The dual solver used in an earlier pass did not converge and understated the score by 0.024.

Table 11. Per-rubric QWK for data augmentation without topic-aware preprocessing (mean $\pm$ std over 10 folds). Significance markers compare each method against Baseline: * $p < 0.05$, ** $p < 0.01$ (paired one-sided t -test, improvement); † significant decrease (two-sided $p < 0.05$). No item shows a significant decrease.

Category	Rubric Item	Baseline	CHEF 1:1	RMI ($r=0.3$)
Expression	Grammar (exp1)	0.2583 $\pm$ 0.0161	0.2589 $\pm$ 0.0148	0.2573 $\pm$ 0.0147
	Vocabulary (exp2)	0.3205 $\pm$ 0.0136	0.3212 $\pm$ 0.0156	0.3262 $\pm$ 0.0180*
	Sentence (exp3)	0.9353 $\pm$ 0.0048	0.9380 $\pm$ 0.0040*	0.9364 $\pm$ 0.0030
Organization	Consistency (org1)	0.3477 $\pm$ 0.0130	0.3564 $\pm$ 0.0131*	0.3561 $\pm$ 0.0137**
	Length (org2)	0.9473 $\pm$ 0.0046	0.9494 $\pm$ 0.0033*	0.9492 $\pm$ 0.0035*
	In-paragraph (org3)	0.9122 $\pm$ 0.0033	0.9140 $\pm$ 0.0029*	0.9135 $\pm$ 0.0029
	Structure (org4)	0.6199 $\pm$ 0.0159	0.6249 $\pm$ 0.0131	0.6210 $\pm$ 0.0123
Content	Clarity (con1)	0.3780 $\pm$ 0.0212	0.3754 $\pm$ 0.0182	0.3797 $\pm$ 0.0277
	Novelty (con2)	0.1428 $\pm$ 0.0153	0.1786 $\pm$ 0.0149**	0.1742 $\pm$ 0.0180**
	Prompt (con3)	0.9490 $\pm$ 0.0033	0.9507 $\pm$ 0.0027**	0.9495 $\pm$ 0.0028
	Descriptions (con4)	0.4481 $\pm$ 0.0153	0.4479 $\pm$ 0.0141	0.4506 $\pm$ 0.0181
Average QWK		0.5690 $\pm$ 0.0078	0.5741 $\pm$ 0.0059*	0.5740 $\pm$ 0.0065**
FCS		0.4243 $\pm$ 0.0120	0.4318 $\pm$ 0.0115**	0.4291 $\pm$ 0.0133*

Table 12. Per-rubric QWK for topic-aware preprocessing (mean $\pm$ std over 10 folds). Significance markers compare Baseline+Topic against Baseline: * $p < 0.05$, ** $p < 0.01$ (paired one-sided t -test, improvement); † significant decrease (two-sided $p < 0.05$). No item shows a significant decrease.

Category	Rubric Item	Baseline	Baseline + Topic
Expression	Grammar (exp1)	0.2583 $\pm$ 0.0161	0.2594 $\pm$ 0.0184
	Vocabulary (exp2)	0.3205 $\pm$ 0.0136	0.3347 $\pm$ 0.0129**
	Sentence (exp3)	0.9353 $\pm$ 0.0048	0.9568 $\pm$ 0.0021**
Organization	Consistency (org1)	0.3477 $\pm$ 0.0130	0.3512 $\pm$ 0.0088
	Length (org2)	0.9473 $\pm$ 0.0046	0.9616 $\pm$ 0.0015**
	In-paragraph (org3)	0.9122 $\pm$ 0.0033	0.9271 $\pm$ 0.0020**
	Structure (org4)	0.6199 $\pm$ 0.0159	0.6278 $\pm$ 0.0113*
Content	Clarity (con1)	0.3780 $\pm$ 0.0212	0.3794 $\pm$ 0.0262
	Novelty (con2)	0.1428 $\pm$ 0.0153	0.1458 $\pm$ 0.0249
	Prompt (con3)	0.9490 $\pm$ 0.0033	0.9566 $\pm$ 0.0009**
	Descriptions (con4)	0.4481 $\pm$ 0.0153	0.4441 $\pm$ 0.0114
Average QWK		0.5690 $\pm$ 0.0078	0.5768 $\pm$ 0.0069**
FCS		0.4243 $\pm$ 0.0120	0.4311 $\pm$ 0.0124**

of the gains is itself informative: the improvements are largest on form-related items (Sentence +0.0215, In-paragraph +0.0149, Length +0.0143) rather than on the content items that topical relevance would most directly concern (Clarity +0.0014, Novelty +0.0030, Descriptions −0.0040, none significant). This pattern is consistent with the topic label acting partly as a per-prompt calibration of the score distribution—each prompt has its own score profile, and all prompts are seen in training under our prompt-stratified protocol—rather than purely as a measure of topical alignment. We report the effect as measured and flag this interpretation as the one

Table 13. Per-rubric QWK for data augmentation on top of topic-aware preprocessing (mean $\pm$ std over 10 folds). Significance markers compare each augmented configuration against Baseline+Topic: * $p < 0.05$, ** $p < 0.01$ (paired one-sided t -test, improvement); † significant decrease (two-sided $p < 0.05$). No item shows a significant decrease.

Category	Rubric Item	Baseline + Topic	CHEF 1:1 + Topic	RMI ($r=0.3$) + Topic
Expression	Grammar (exp1)	0.2594 $\pm$ 0.0184	0.2622 $\pm$ 0.0185	0.2603 $\pm$ 0.0160
	Vocabulary (exp2)	0.3347 $\pm$ 0.0129	0.3357 $\pm$ 0.0157	0.3329 $\pm$ 0.0168
	Sentence (exp3)	0.9568 $\pm$ 0.0021	0.9572 $\pm$ 0.0013	0.9571 $\pm$ 0.0018
Organization	Consistency (org1)	0.3512 $\pm$ 0.0088	0.3573 $\pm$ 0.0105*	0.3546 $\pm$ 0.0128
	Length (org2)	0.9616 $\pm$ 0.0015	0.9618 $\pm$ 0.0013	0.9618 $\pm$ 0.0015
	In-paragraph (org3)	0.9271 $\pm$ 0.0020	0.9274 $\pm$ 0.0017	0.9271 $\pm$ 0.0018
	Structure (org4)	0.6278 $\pm$ 0.0113	0.6265 $\pm$ 0.0126	0.6291 $\pm$ 0.0156
Content	Clarity (con1)	0.3794 $\pm$ 0.0262	0.3842 $\pm$ 0.0232	0.3796 $\pm$ 0.0251
	Novelty (con2)	0.1458 $\pm$ 0.0249	0.1717 $\pm$ 0.0232**	0.1732 $\pm$ 0.0214*
	Prompt (con3)	0.9566 $\pm$ 0.0009	0.9573 $\pm$ 0.0012*	0.9575 $\pm$ 0.0009**
	Descriptions (con4)	0.4441 $\pm$ 0.0114	0.4483 $\pm$ 0.0127	0.4509 $\pm$ 0.0120*
Average QWK		0.5768 $\pm$ 0.0069	0.5809 $\pm$ 0.0067*	0.5804 $\pm$ 0.0063**
FCS		0.4311 $\pm$ 0.0124	0.4360 $\pm$ 0.0122*	0.4347 $\pm$ 0.0134*

a reader should keep in mind; separating the two mechanisms would require a prompt-held-out protocol, which our design does not provide (Section 4.3).

Augmentation on top of topic-aware preprocessing is consistently positive, but not established. To formally address whether the combination provides additional benefits, we treat Baseline+Topic as the reference and test the augmented configurations against it. CHEF+Topic reaches an FCS QWK of 0.4360 ($\Delta = +0.0048$, $p = 0.0448$, 7/10 folds) and RMI+Topic reaches 0.4347 ($\Delta = +0.0035$, $p = 0.0464$, 7/10 folds); the average rubric QWK gains are likewise significant ($p = 0.0434$ and $p = 0.0075$, respectively). The additional gains are again driven primarily by the content items—Novelty (CHEF+T $p = 0.0066$; RMI+T $p = 0.0309$), Prompt ($p = 0.0195$ / $p = 0.0011$), and Descriptions (RMI+T $p = 0.0329$)—and no item is significantly degraded (Table 13). The direction is therefore consistent: the best configuration in this study combines both components. The strength of this particular claim should not be overstated, however. At the FCS level the two comparisons are nominally significant (0.0448 and 0.0464) but do *not* survive the Holm adjustment over the five confirmatory comparisons (adjusted 0.0895 for both), and the distribution-free Wilcoxon test agrees ($p = 0.065$ and $p = 0.080$). Under the criterion declared in Section 5.1 we therefore report the incremental effect of augmentation on top of topic conditioning as *suggestive rather than established*: the effect is positive in 7 of 10 folds in both cases and significant at the average-rubric level, but a single-seed, ten-fold experiment of this size cannot settle a difference of 0.004 QWK. The three comparisons that do survive adjustment—each component against the plain baseline—are the results we regard as established.

5.5 Sensitivity to Augmentation Ratio

The main results above (Tables 11–13) adopt a 1:1 augmentation ratio—one augmented essay per original—for both CHEF and RMI. To verify that this choice is not arbitrary, we swept the ratio over $\{0.5, 1.0, 1.5, 2.0, 2.5\}$ augmented copies per original essay, holding every other setting fixed, under both the plain and topic-aware conditions and under the same 10-fold protocol. Rather than tabulating all 20 method $\times$ condition $\times$ ratio cells—which would largely restate, at finer

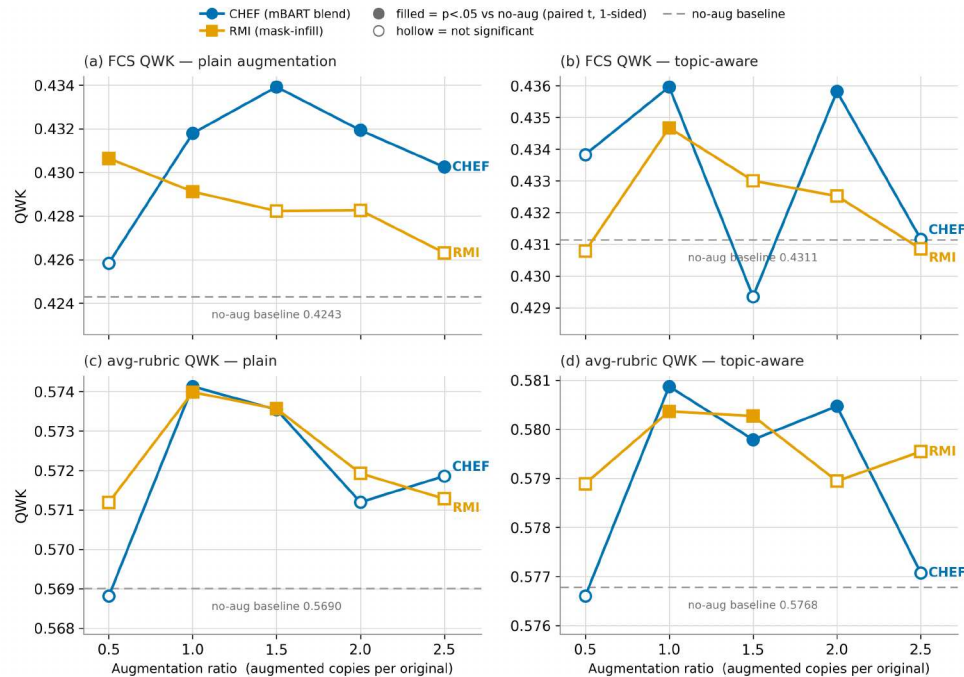


Figure 5. FCS QWK (top) and average rubric QWK (bottom) across augmentation ratios for CHEF and RMI, under the plain (left) and topic-aware (right) conditions. Filled markers denote $p < 0.05$ against the corresponding no-augmentation reference (paired one-sided t -test); hollow markers are not significant. The dashed line marks the no-augmentation reference.

granularity, the same FCS and rubric-level quantities already reported in Tables 11–13—we report this sweep as a single figure (Figure 5) and summarize only the findings that bear on the choice of ratio.

Three patterns emerge. First, a 0.5:1 ratio is not a usable operating point. In three of the four method $\times$ condition combinations it produces no significant FCS gain (CHEF plain $p = 0.356$; CHEF topic-aware $p = 0.155$; RMI topic-aware $p = 0.554$), and under the topic-aware condition the RMI point estimate is even slightly negative ($\Delta = -0.0004$). The single exception—RMI in the plain condition ($\Delta = +0.0063$, $p = 0.0074$)—does not carry over to the rubric level: its average rubric QWK is unchanged ($p = 0.134$) and one item, *Descriptions* (con4), degrades significantly (-0.0033 , $p = 0.027$). Halving the augmented material therefore buys either nothing or an unstable gain.

Second, gains do not increase monotonically with the ratio, and ratio 1.0 is the only point at which all four method $\times$ condition combinations are significant ($p = 0.0044$, 0.0106, 0.0448 and 0.0464; the same four comparisons as rows 1, 2, 4 and 5 of Table 10). Beyond it the behaviour becomes erratic rather than better. CHEF attains its highest plain-condition score at ratio 1.5 ($\Delta = +0.0096$, $p = 0.0010$), but at that same ratio falls below the reference in the topic-aware condition ($\Delta = -0.0018$, $p = 0.790$), and at ratio 2.5 returns to it exactly ($\Delta = +0.0000$, $p = 0.497$). RMI’s gains taper off above ratio 1.0 and lose significance in both conditions (plain: $p = 0.064$, 0.084, 0.081; topic-aware: $p = 0.218$, 0.302, 0.558 at ratios 1.5, 2.0 and 2.5). Third, every significant rubric-level degradation observed in the sweep occurs away from ratio 1.0: RMI in the plain condition degrades *Descriptions* at ratios 0.5 and 1.5 and *Clarity* at ratio 2.5, and CHEF in the topic-aware condition degrades *In-paragraph* at ratio 2.5, whereas no configuration at ratio 1.0 degrades any item.

In the topic-aware condition—which produces the best configuration reported in Table 10—ratio 1.0 is the highest-scoring point for CHEF (FCS QWK 0.4360) among all five ratios tested, ahead of ratio 2.0 (0.4358) by a margin far smaller than the fold-level standard deviation. We therefore adopt 1:1 as a single operating ratio for both methods throughout the paper: it is the only ratio significant in all four method×condition combinations, the only one that degrades no rubric item anywhere in the sweep, and the one that coincides with the best-performing setting precisely in the topic-aware condition that constitutes our final configuration—while also being the least data-hungry of the ratios that work, doubling the training set rather than tripling or quadrupling it. Ratios beyond 1.5 provide no additional benefit, and at the highest ratio tested (2.5) the topic-aware condition loses significance for both methods—consistent with training data becoming increasingly dominated by generator-introduced noise as the oversampling factor grows.

5.6 Discussion: Interpreting Absolute QWK Against Human Agreement

The absolute FCS QWK values reported above (0.42–0.44) may appear low compared to figures around 0.8 sometimes reported in the AES literature. Two factors explain this gap. First, published QWK values are generally not comparable across studies of this corpus: prediction targets, score scales, binning schemes, and validation protocols differ substantially (Section 2.2), and QWK is highly sensitive to the number of classes and their distribution. Second, and more fundamentally, the annotation-quality analysis of Section 3.3 establishes what is attainable on this metric at all. Under the identical FCS quartile-binning procedure, a human rater scored against the average of the other two—the same target the model is evaluated against—reaches a QWK of 0.523. Our baseline (0.4243) and best configuration (0.4360) thus stand at 81% and 83% of the human ceiling for this exact protocol. The absolute values are low because the ceiling itself is low, not because the models fall far short of what human raters achieve; but they do fall short of it, and we do not claim parity.

The analysis also reframes the per-rubric performance differences. The model’s per-item QWK profile tracks the human agreement profile closely (Spearman $\rho = 0.818$ against the LOO reference; Pearson $r = 0.951$, inflated by the two-cluster structure of the items); the items on which the model scores lowest (Novelty, 0.1428; Grammar, 0.2583) are precisely those on which human raters disagree most (LOO agreement 0.295 and 0.387; Table 5), and on six items the model meets or exceeds the agreement a human rater reaches with the average of the other two. This indicates that the remaining errors on subjective items largely reflect label noise rather than correctable model deficiencies. It is particularly noteworthy that the largest augmentation gains arise exactly on Novelty, the noisiest item: augmentation appears to compensate for data scarcity where the learning signal is weakest, although improvements on items with low human agreement should be interpreted with additional caution, as QWK estimates are inherently noisier there. The composite-level picture is different from the item-level one: there the model remains clearly below the human ceiling (Section 3.3), and the gap of roughly 0.10 QWK is the margin that future work on this corpus has to close.

6. Ethical Considerations

Automated essay scoring is a high-stakes application: the essays in our corpus were written by students from grade 4 through grade 12, and scoring outcomes can influence the educational trajectories of minors. We therefore discuss the ethical implications of this work and the precautions we consider necessary for any deployment.

Human-in-the-loop deployment. Given the performance characteristics documented in this paper, we do not consider fully automated, unassisted scoring to be an appropriate use of the presented system. Its intended role is to support human raters—for example, as a second opinion,

a consistency check, or a triage mechanism that flags essays for closer human review. Final scoring authority in any consequential setting should remain with trained human raters.

Rubric-level reliability stratification. The inter-annotator agreement analysis in Section 3.3 provides a principled basis for deciding *which* automated judgments can be trusted. Rubric items with near-ceiling human agreement and model performance (Sentence, Length, In-paragraph, Prompt; QWK ≥ 0.91) are plausible candidates for automated assistance. Conversely, for subjective items with low human agreement (Novelty, IAA 0.284; Clarity, 0.339; Descriptions, 0.364), neither human nor model scores are reliable in isolation; automated scores on these items should be presented with explicit uncertainty indications, if at all, and should never be used as the sole basis for feedback or grading. Communicating this stratification to teachers and students is, in our view, a precondition for responsible use.

Fairness and shortcut risks. Essay length is strongly correlated with school level in this corpus (Section 3.1), and length-related rubric items are among the easiest to predict. This creates a risk that models exploit length as a shortcut, potentially disadvantaging concise writers or students whose length norms differ from their cohort. Deployments should monitor performance separately across school levels and demographic groups where lawful and appropriate, and should audit for length-driven scoring artifacts.

Data privacy. All experiments use the *Essay Writing Evaluation Data*, which was collected, de-identified, and publicly released by the National Information Society Agency (NIA) through AI-HUB under its own consent and privacy framework. No new personal data were collected for this study, and augmented essays are derived exclusively from the de-identified public corpus.

Explainability. The system presented here produces a number for each rubric item and no account of it: the essay is represented as a sequence of frozen sentence embeddings compressed by a GRU, and nothing in the architecture identifies which passage of an essay drove a given score. For a scoring decision that a student may wish to contest, this is a substantive shortcoming, and it is a second reason—alongside the reliability limits above—to restrict the system to a human-assisted role. Interpretable scoring has been pursued in the Korean AES literature through rubric-aligned models and tree-based scorers whose features can be inspected (Lee et al. 2023; im Park and ki Choi 2023), and more recently through LLM-based trait-level scoring that emits a rationale alongside the score (Lee et al. 2024). Predicting all eleven criteria separately, as we do, is a partial step in this direction—a per-criterion score is more actionable to a teacher than a single holistic number, and the stratification above indicates which of those criteria to trust—but it is not an explanation. Combining linguistically informed augmentation with an explicit explanation mechanism is, in our view, the most valuable direction for making systems of this kind deployable in classrooms, and we leave it to future work.

Transparency and reproducibility. To support scrutiny of the claims made here, we report exact dataset construction steps, fold splits, statistical procedures, augmentation filtering criteria, and rubric weights in full. We believe such transparency is particularly important for systems that may inform the assessment of students.

7. Conclusion

This study advances automated essay scoring for the Korean language by systematically evaluating two linguistically informed data augmentation techniques—CHEF and Random Masking Insertion—together with a topic-aware preprocessing strategy, under a statistically rigorous protocol. Using 10-fold prompt-stratified cross-validation with paired significance tests on 44,658 essays, we showed that CHEF (1:1) and RMI ($r = 0.3$, 1:1) each significantly improve scoring performance and that topic-aware preprocessing contributes an independent significant gain, all three surviving a multiplicity correction. Augmentation applied on top of topic-aware preprocessing improves the composite-score QWK further, from 0.4243 to 0.4360 for the best configuration

and with no significant degradation on any rubric item, although this last increment is nominally rather than adjusted-significant and we treat it as suggestive. The gains concentrate on the most difficult content criteria, most notably *novelty*, where the baseline is weakest.

Beyond these performance results, this study contributes an empirical characterization of the corpus itself. By recovering the three raters' original scores, we quantified human agreement for every rubric item under the protocol used to score the model, comparing each rater against the average of the other two—the same target the model predicts. The model's per-item performance profile closely tracks this profile (Spearman $\rho = 0.82$), and the composite-score human ceiling is 0.523. This analysis (Section 5.6) reframes absolute QWK values on this corpus: values near 0.43 are not evidence of a weak model but of a metric whose human ceiling is itself close to 0.5, although the model does remain measurably below that ceiling, and the remaining errors on subjective items largely reflect annotation noise rather than correctable model deficiencies. A supplementary comparison with the official word2vec-based reference models distributed with the corpus and five traditional machine learning methods provides additional reference points for these results.

A ratio-sensitivity analysis over five augmentation levels (Section 5.5) confirmed that this 1:1 configuration—for both CHEF and RMI ($r = 0.3$)—is not an arbitrary choice: gains are non-monotonic in the ratio, a 0.5:1 ratio is insufficient in three of the four settings tested and unstable in the fourth, and ratios beyond 1.5:1 provide no further benefit. Ratio 1.0 is in fact the only level at which every method and condition improves significantly and no rubric item degrades, so the simplest configuration that avoids oversampling the training set is also the most reliable one, providing practical guidance for future Korean AES development. Future research directions include exploring adaptive augmentation strategies responsive to specific prompts or rubric items, integrating more advanced pre-trained language models, obtaining human judgments of grade-level plausibility for augmented text, and developing ensemble approaches to further boost AES performance. We hope that the combination of linguistically grounded augmentation, topic-aware preprocessing, and annotation-aware evaluation presented here provides a solid and transparent foundation for subsequent work on Korean AES.

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